# CS 1710 Process Book

Google Docs Link

**Milestone 1 | Team Formation**

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## Milestone 2 | Map

**Topic**

- Post-graduation plans for early career professionals, especially regarding popular cities for relocation, salary, and housing.

**Audience**

I. Initial Discussion

- Our audience would focus on students pursuing higher education, who often have to make plans to relocate for work or academic reasons. Three potential options for a target audience are:

    1. **Undergraduate students at the upperclassmen level:** These individuals are more likely to be thinking about moving for work related reasons. Including juniors into our target audience ensures our visualization tool can also help students hoping to turn internships into full-time offers.

    2. **College and career center visitors:** Keeping our target audience broader means students and the professionals who advise them are able to use our visualization tool.

    3. **Professional student organizations:** Instead of building for the general student body, narrowing down to students in organizations already geared toward professional pathways means we are reaching the exact students who would benefit from our visualization.

II. Target Audience

- We selected undergraduate students at the upperclassmen level to be our target audience. We felt this group kept a good balance between broad and specific. By considering all students at this level, our visualization is able to guide students regardless of their career pursuits. This ensures we cover pathways like academia, service projects, creative roles, and more. Since we are building for upperclassmen, we assume that they are technologically literate and comfortable with tools like websites and dashboards, as long as we are also clear about how to use them. They also know how to critically think through major decisions like salaries, cost of living, and job opportunities, but their interest may be in comparing these factors across different places. As for the level of detail, we might provide brief descriptions for geographic areas, but be more specific in the numbers, such as average rent, gas prices, and taxes.

## III. The Questions

- The questions our audience might have about our early-career data are:
    1. What starting salary would someone need to live in this specific place?
    2. How can I compare cities by the average starting salary of their younger residents?
    3. Which areas of the United States have the lowest cost of living?
    4. Which U.S. states pay the most in taxes?
    5. Where does an entry-level salary go furthest after adjusting for cost of living?
    6. How much of a recent graduate's monthly income would go toward rent in each city?
    7. Which cities have the most job opportunities for recent graduates?
    8. Which cities have the largest populations of people in their twenties?
    9. Where could a recent graduate realistically afford to live without roommates?
    10. How much could having roommates reduce living expenses in different cities?
    11. Which cities are easiest to navigate without owning a car?
    12. How do average commute times differ across cities?
    13. Which cities offer the best balance between career opportunities and affordability?
    14. Which cities have become more or less affordable for recent graduates over time?
    15. What expenses besides rent account for the largest differences in cost of living?
    16. Which less-popular cities provide similar opportunities to major cities at a lower cost?
    17. How does the best city change depending on someone's salary, industry, and lifestyle priorities?
    18. How much could a recent graduate expect to save each month in different cities?

## IV. The Data

We plan to collect data on major U.S. cities that recent college graduates may consider moving to. Each observation will represent a city and include:

- City and state: categorical and geographic
- Region of the United States: categorical
- Population and population aged 20-29: quantitative
- Average starting salary for younger workers: quantitative

- Employment opportunities by industry: categorical and quantitative
- Overall cost-of-living index: quantitative
- Median monthly rent: quantitative
- Average costs of food, transportation, and utilities: quantitative
- State and local income-tax rates: quantitative
- Estimated after-tax income: quantitative
- Average commute time: quantitative
- Public-transit availability: ordinal or quantitative
- Walkability: quantitative score
- Typical number of roommates: ordinal
- Estimated monthly savings or disposable income: quantitative

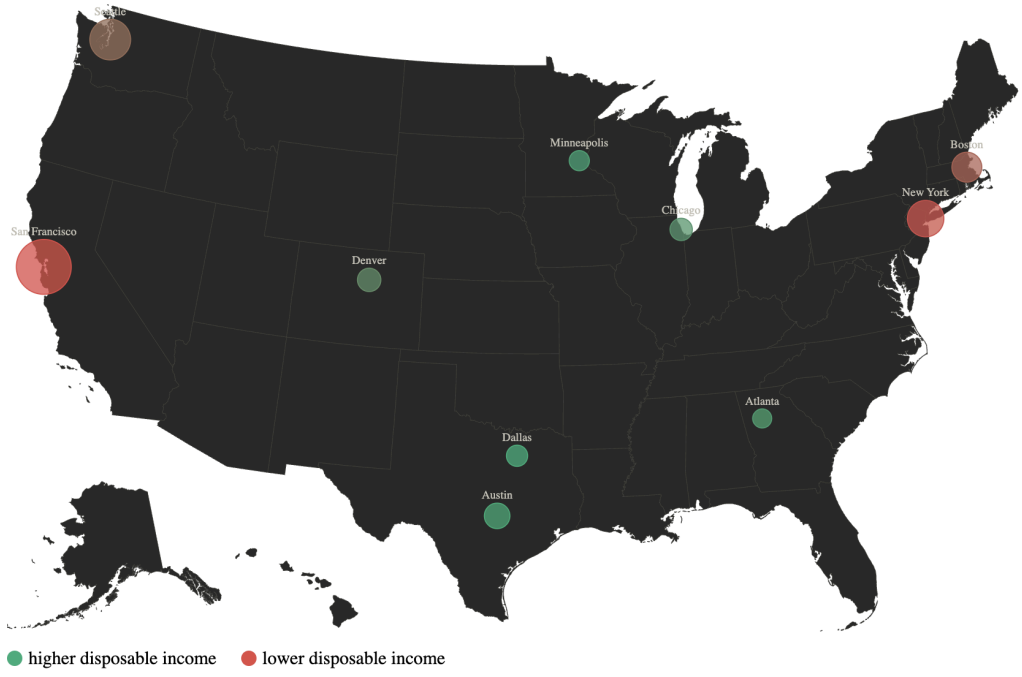
We would likely draw from sources such as the U.S. Census Bureau, Bureau of Labor Statistics, housing datasets, and cost-of-living databases. We may also derive new variables, such as rent as a percentage of starting salary and estimated disposable income, to make comparisons between cities more meaningful.

## GROUP VISUALIZATIONS

Anna Feng

**#1: "Where Can My Salary Take Me?" (U.S. bubble map: bubble size = starting salary, bubble color = disposable income) made via Claude**

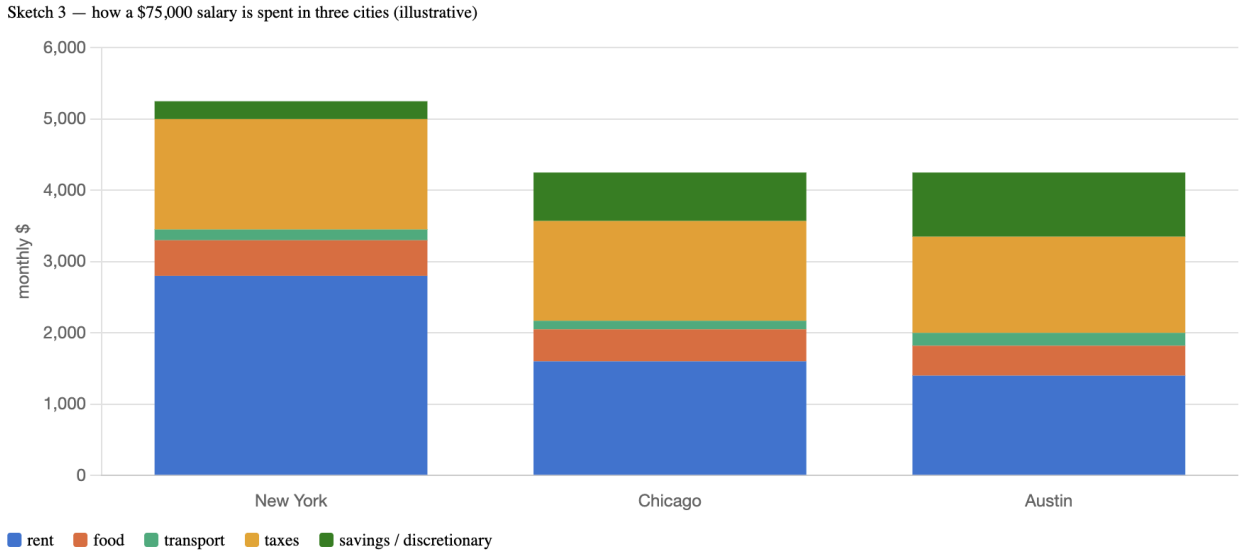
Sketch 1 — bubble size = starting salary, color = estimated disposable income (illustrative placeholder data)




**#2: "Salary vs. Cost of Living" (scatterplot: cost-of-living index vs. starting salary) made via Claude and GPT**



**#3: "Where Does My Paycheck Go?" (stacked budget comparison across cities) made via Excel**




Reflection: My sketches build on the team's original questions but sharpen them around the actual decision I believe our audience faces: where to live after college. The original questions treated salary, cost of living, and taxes as separate axes worth understanding individually. Once we started sketching, it is clearer that no single one of those metrics answers the question a graduating student actually asks, which is closer to "what would my life feel like there?" Sketch 1 keeps salary as a headline number but makes it more insightful by layering in disposable income, so a high-salary city can visibly read as a worse deal. Sketch 2 makes the salary-vs-cost of living tradeoff more explicit, which our original separately brainstormed questions never forced us to confront directly. Sketch 3 goes a step further and breaks the paycheck into categories, showing that "disposable income" is itself a vague concept. The really valuable information is in the shape of the budget, not just the leftover total. I deliberately set aside the transit, social-scene, and distance-from-home questions for these initial visualizations, not because they are less interesting, but because they are harder to quantify consistently across 15-20 cities, and we wanted our first sketches to prove out the income/cost core before layering in softer, harder-to-source variables. I don't think the original questions were wrong—rent, salary, and taxes are still the inputs—but I now believe the compound question ("what does this salary buy me here, after the basics?") is more useful to a reader than any of the original questions in isolation, because it's the version a person can actually act on (i.e., what I would actually find interesting and actionable as a user!).

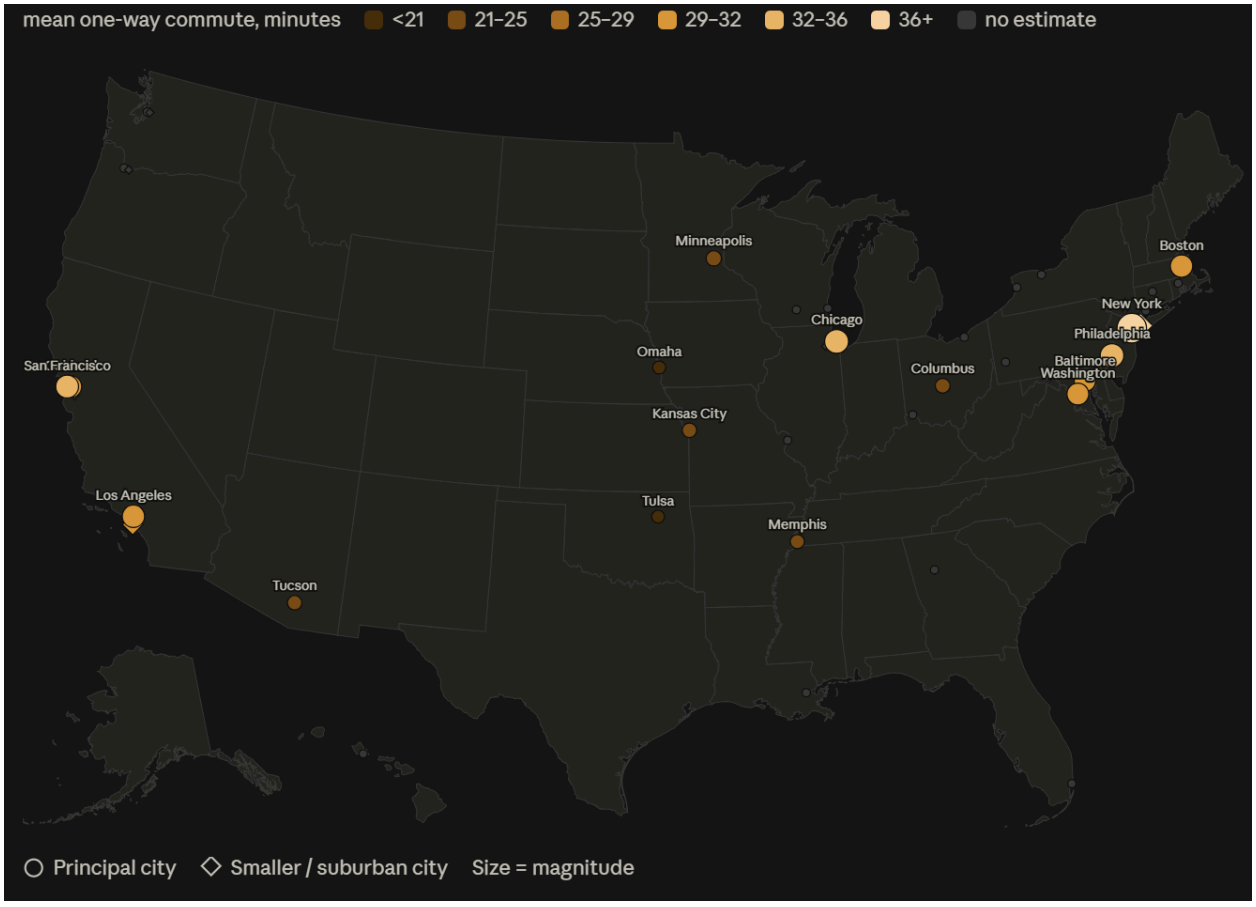
Additional Considerations (Divergence Log):

- Commute/transit lens: score cities on transit access and average commute time, independent of money, for students who weigh quality-of-life over optimization.

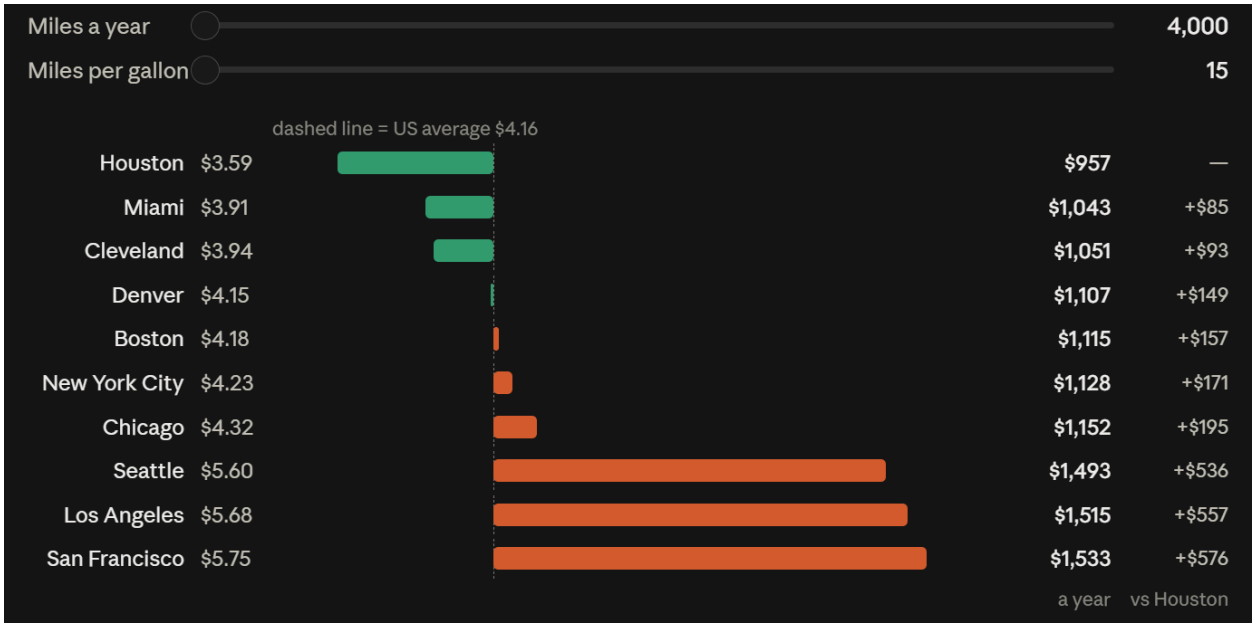
- "Distance from home" ring map: concentric bands radiating from the user's hometown, showing which cities fall within a 2-hour flight vs. a full day of travel.
- Career-density heatmap: a choropleth by industry showing where entry-level jobs in a given field actually concentrate. None of our three sketches directly visualize this, even though "employment opportunities" was one of our original factors.
- Age/social-scene map: bubble map sized by share of population aged 22–29, addressing the "concentration of other young adults" factor that's unaddressed.
- Personalized ranking / slider tool — closer to our original pitch: sliders for affordability, career opportunity, transit, weather, and proximity to home, producing a live-updating ranked list rather than a single fixed chart. Definitely more ambitious, but worth keeping as a longer-term direction.
- 5-year trajectory: instead of a single-year snapshot, a line chart showing how disposable income in each city changes over an imagined 5-year career progression, since starting salary alone warps a city's long-term value.

**Lorena Alvarez**

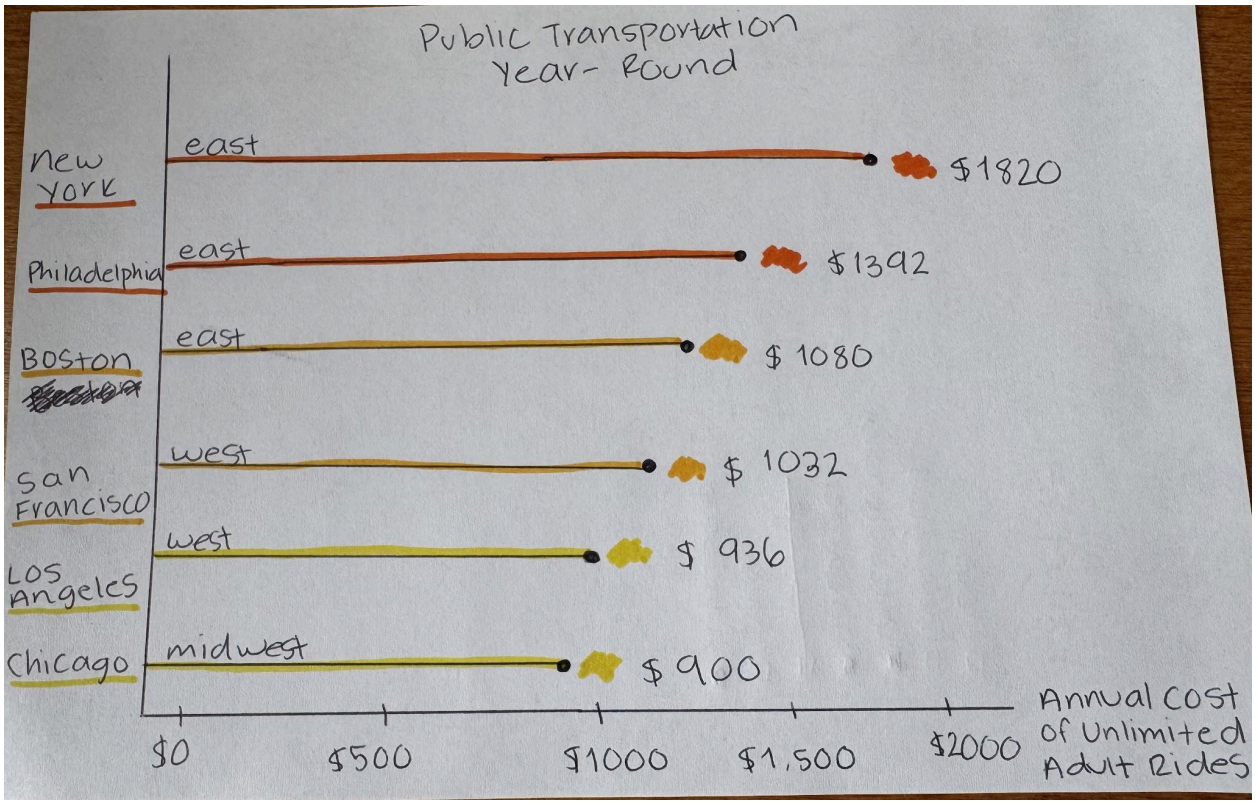
**#1: “How does average commute times differ across cities?” (U.S. bubble map: bubble size = city magnitude, bubble color = mean one-way commute, minutes) made via Claude**



**#2: “How much will I be paying for gas in this city?” (Vertical bar chart with slider interaction) made via Claude**




**#3: “How much will I be paying for public transportation in this city?” (Categorical line chart) made via pen and paper**

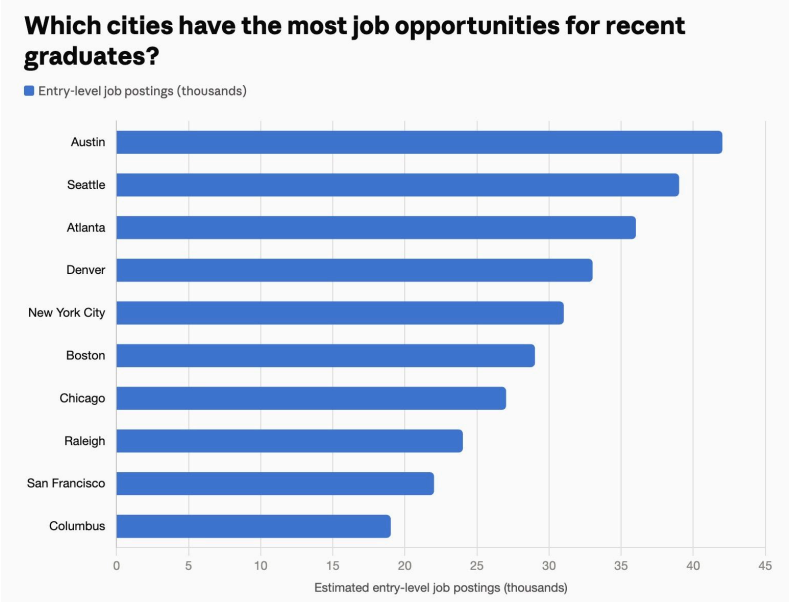


Reflection:

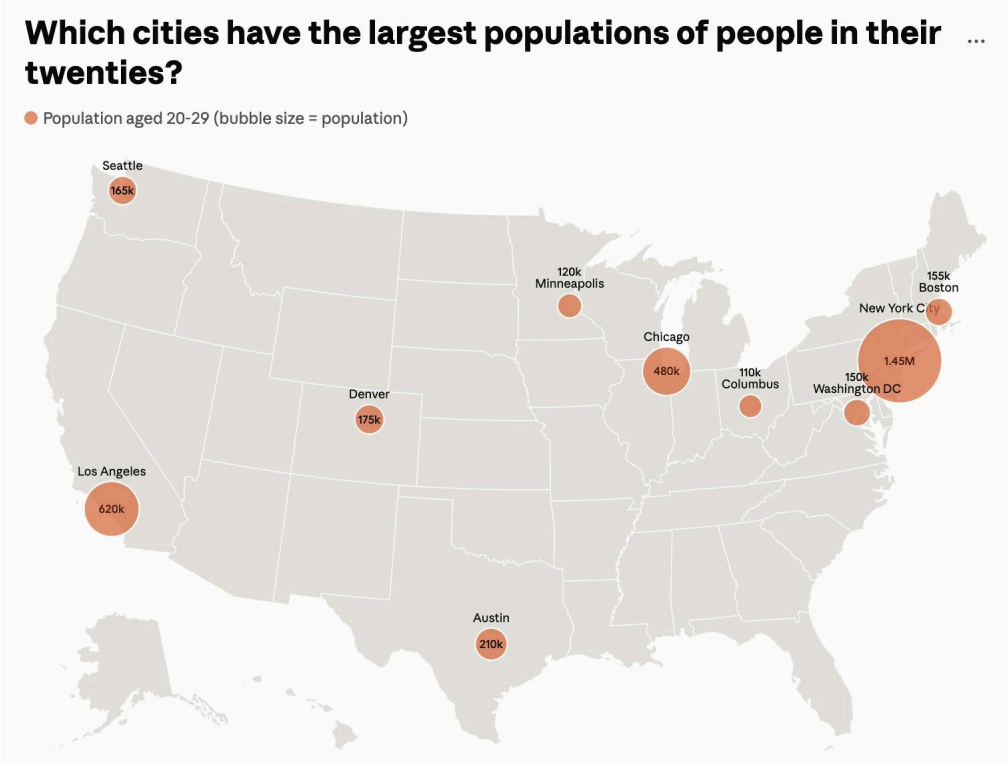
Our sketches use the team's original questions as a guideline, but become more specific to transportation costs across cities, focusing on commute times, gas prices, and subway fare. They differ because they're more realistic questions a newcomer to a city might be asking. They would actually consider buying a car or monthly subway fare subscriptions. These questions are useful because they focus on everyday expenses and experiences that any citygoer will navigate through. However, because they're less idealistic, if we go forward with these questions, the visualization tool might be less helpful towards people just starting to think about relocation. Questions about salary and cost of living like the original questions are more immediate to the mind than gas. These concerns would all be part of our future team discussions about the intent of our visualization and whether it aligns with the audience we are building for. My experience generating visualizations with Claude was interesting because it considered manual input from the user, such as in the #2 Gas Prices sketch. This interactivity could be carried over onto our final visualization tool, where we might have the audience input their prospective salary range and navigate city prices on that baseline.

Ivy Qin:

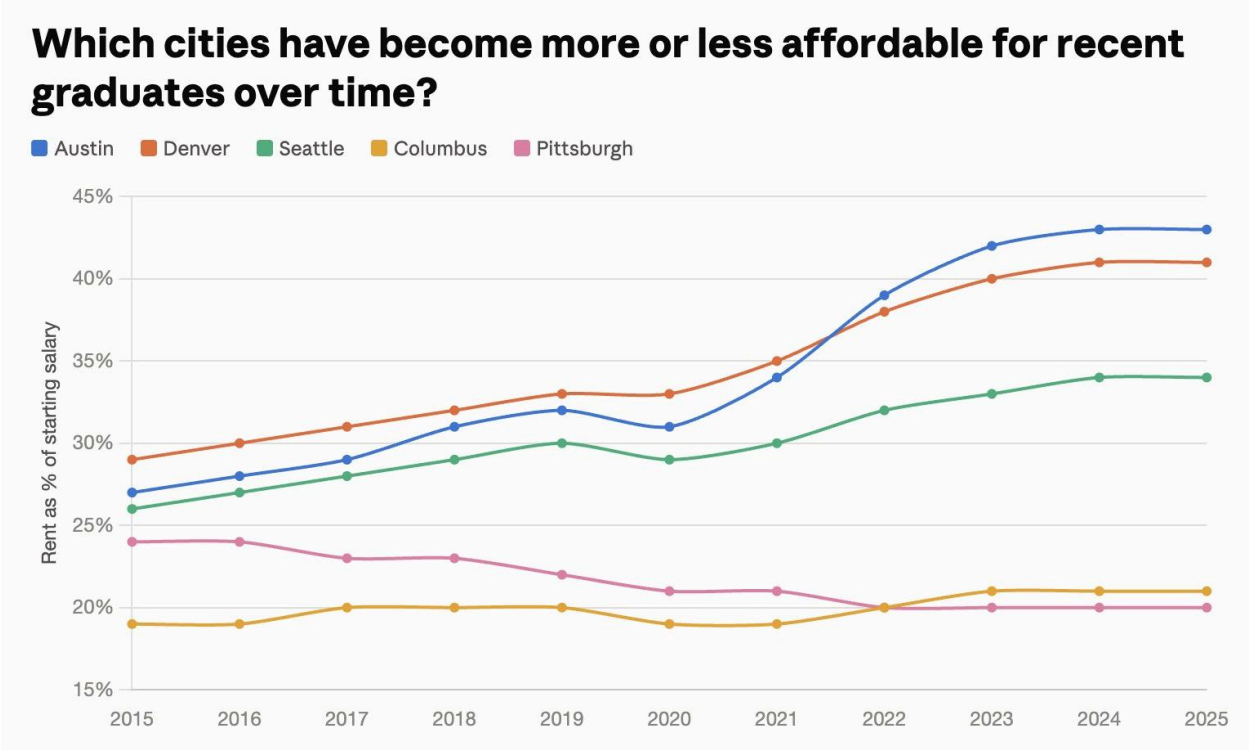
#1: Which cities have the most job opportunities for recent graduates? (ranked bar chart, made by Claude)




#2: Which cities have the largest populations of people in their twenties? (bubble map, made by Claude)




#3: Which cities have become more or less affordable for recent graduates over time? (line chart, made by Claude)

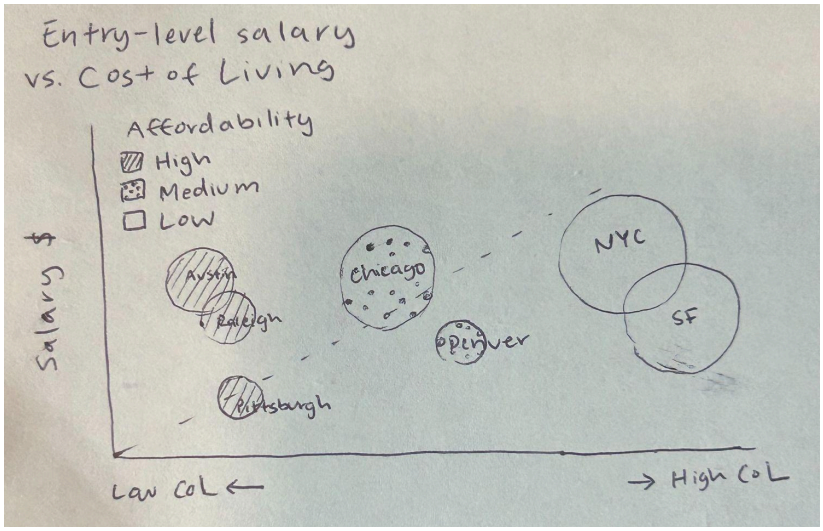



Reflection:

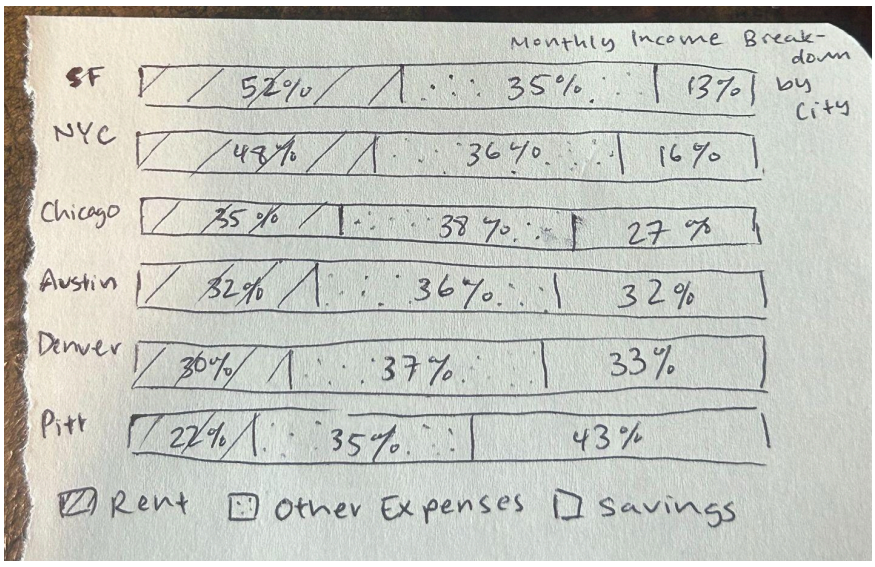
For my visualizations, I chose to focus on job opportunities, social environment, and changes in affordability over time. These questions build on our initial focus of where recent graduates should live, but expand the analysis beyond salary and immediate living expenses. My first visualization addresses which cities have the most job opportunities for recent graduates, since a city's affordability is less useful if it does not offer strong employment prospects. My second visualization looks at which cities have the largest populations of people in their twenties, showing the social environment and concentration of other young adults that could be important to recent graduates in deciding where to move. For the last visualization, I chose to examine how cities have become more or less affordable over time. While the other two visualizations I made give a snapshot of current conditions, this one shows how rent, starting salaries, and disposable income have changed over time, which could reveal trends that are not obvious from a single year of data. These questions differ from some of our team's original questions because I wanted to consider factors beyond the immediate financial tradeoff between salary and cost of living. Although the financial questions are directly relevant for budgeting, I think my questions are also valuable because they provide a broader picture of what makes a city attractive to recent grads including career opportunities, community, and long term affordability.

## Milestone 3 | Sketch

Anna Feng
#1: Entry-level salary vs. cost of living bubble chart
→ Question 5: Where does an entry-level salary go furthest after adjusting for cost of living? Also relates to Question 13 about career opportunities and affordability.
**Sketch ID 1 | Question ID 5**



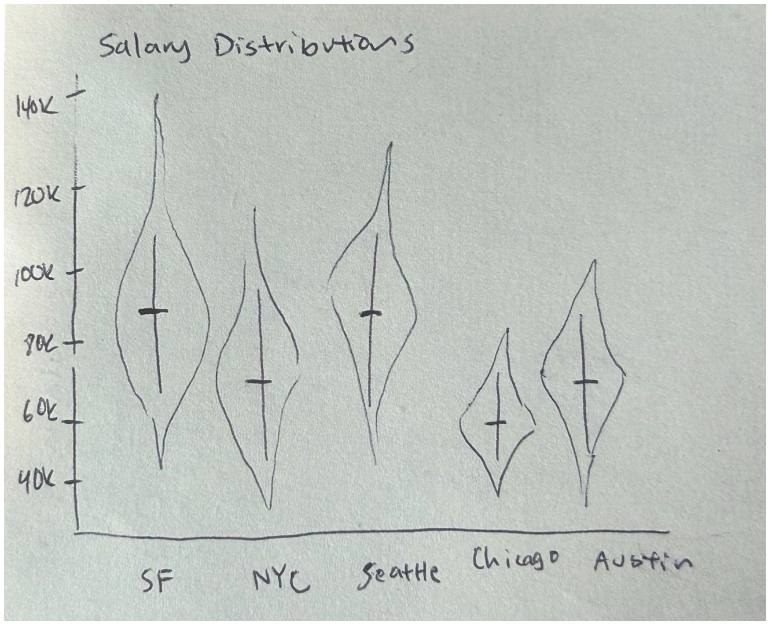

#2: Monthly income breakdown by city
→ Question 18: How much could a recent graduate expect to save each month in different cities? Also answers Question 6 on rent and Question 15 on other expenses.
**Sketch ID 2 | Question ID 18**




#3: Salary distribution violin plots

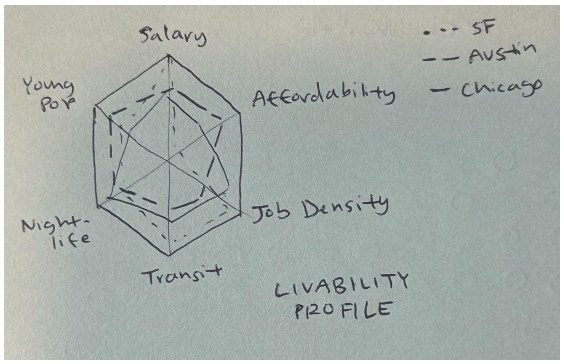
→ Question 2: How can I compare cities by the average starting salary of their younger residents? This is stronger than showing only averages because it also displays the range and distribution of salaries.

**Sketch ID 3 | Question ID 2**



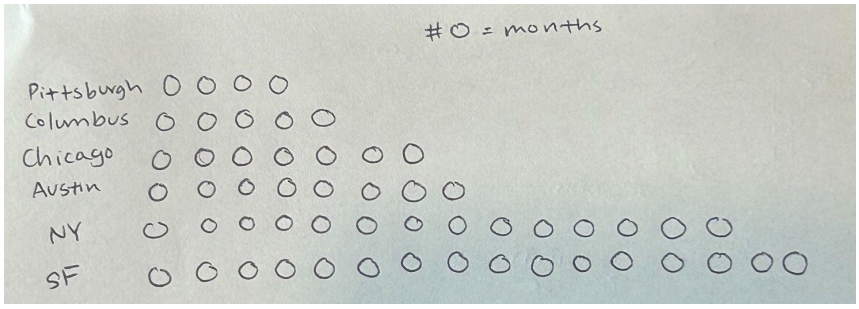

#4: Which city offers the best overall balance of affordability and opportunity?

**Sketch ID 4 | Question ID 4**




#5: How long would it take to reach a savings goal in each city?

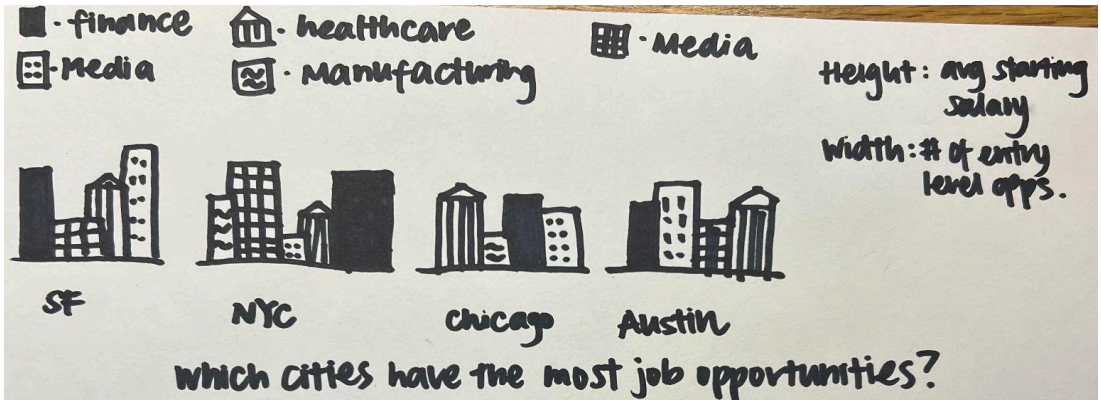
**Sketch ID 5 | Question ID 5**




Ivy Qin

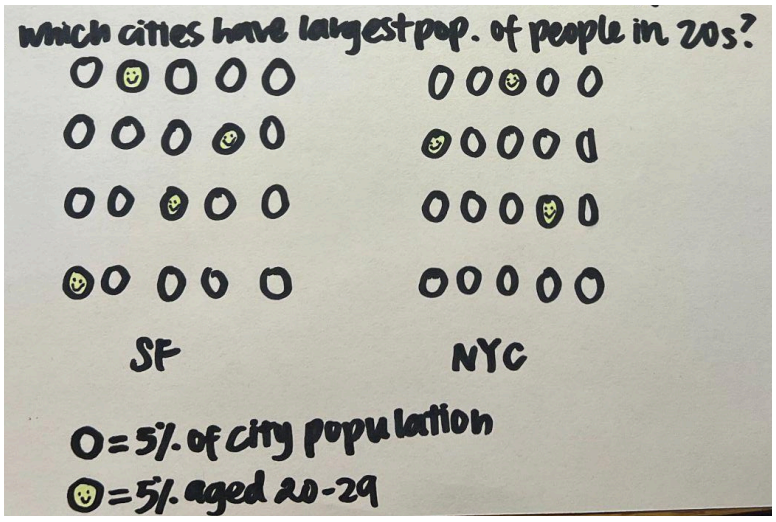
#7: Which cities have the most job opportunities for recent graduates?

**Sketch ID 6 | Question ID 7**



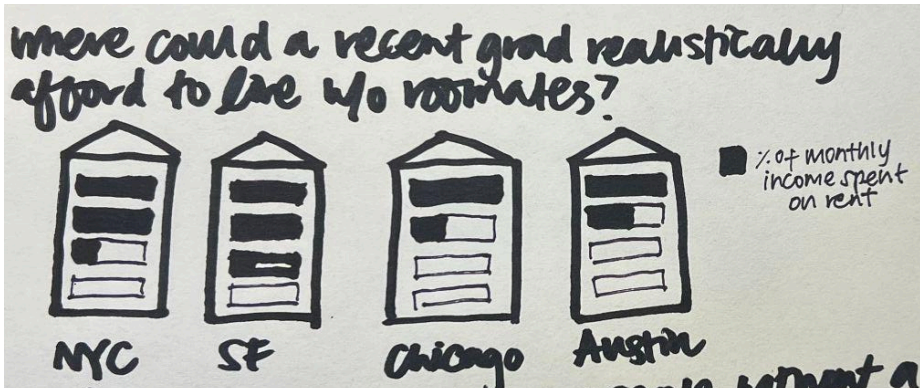

#8: Which cities have the largest populations of people in their twenties?

**Sketch ID 7 | Question ID 8**

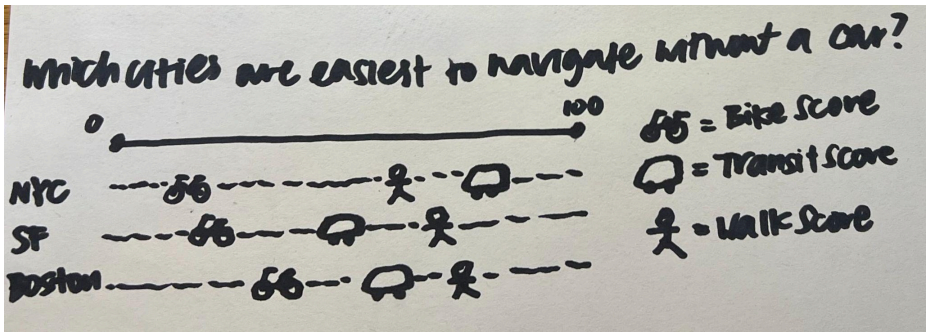



#9: Where could a recent graduate realistically afford to live without roommates?

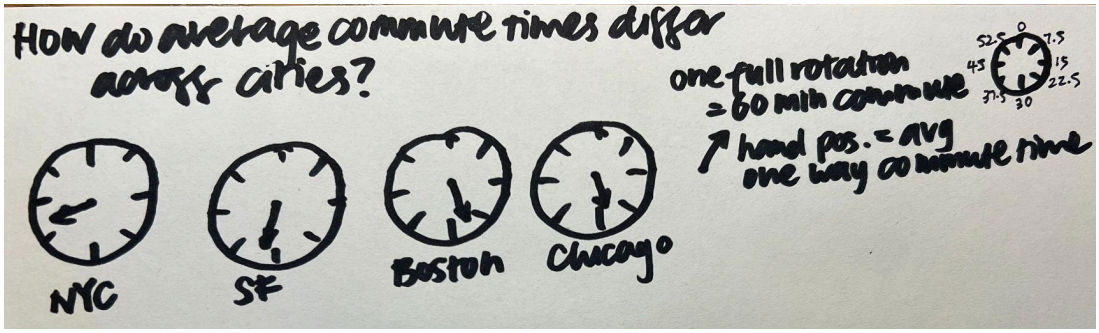
**Sketch ID 8 | Question ID 9**




#11: Which cities are easiest to navigate without owning a car?
**Sketch ID 9 | Question ID 11**

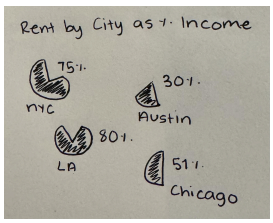



#12: How do average commute times differ across cities?
**Sketch ID 10 | Question ID 12**

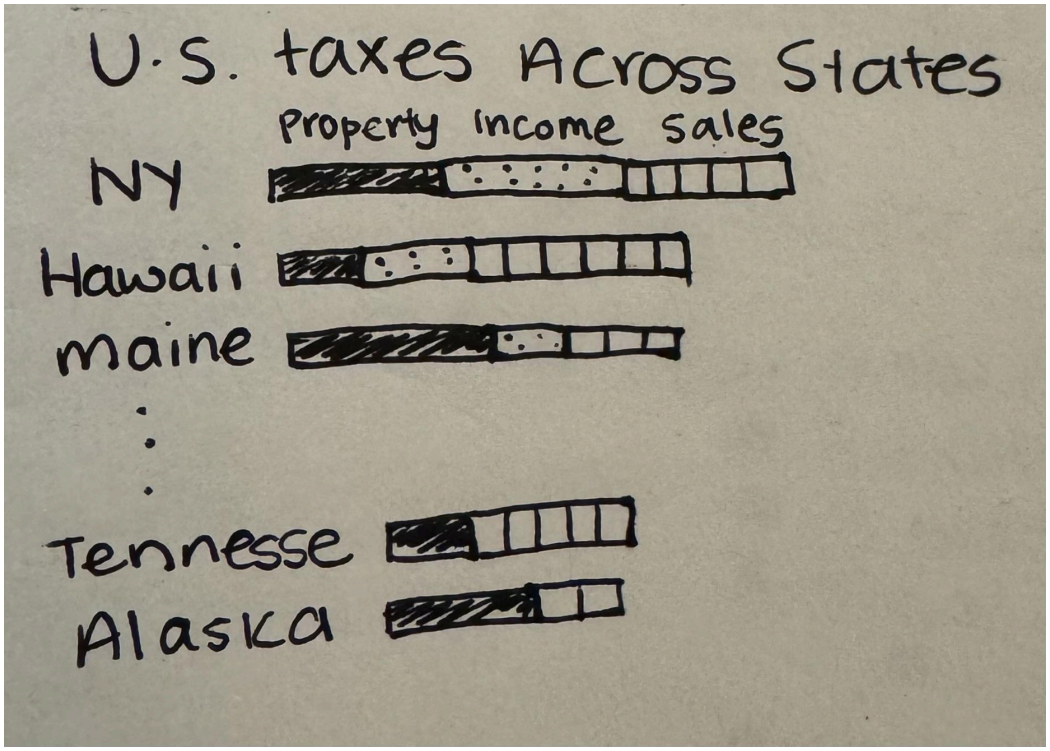



Lorena Alvarez

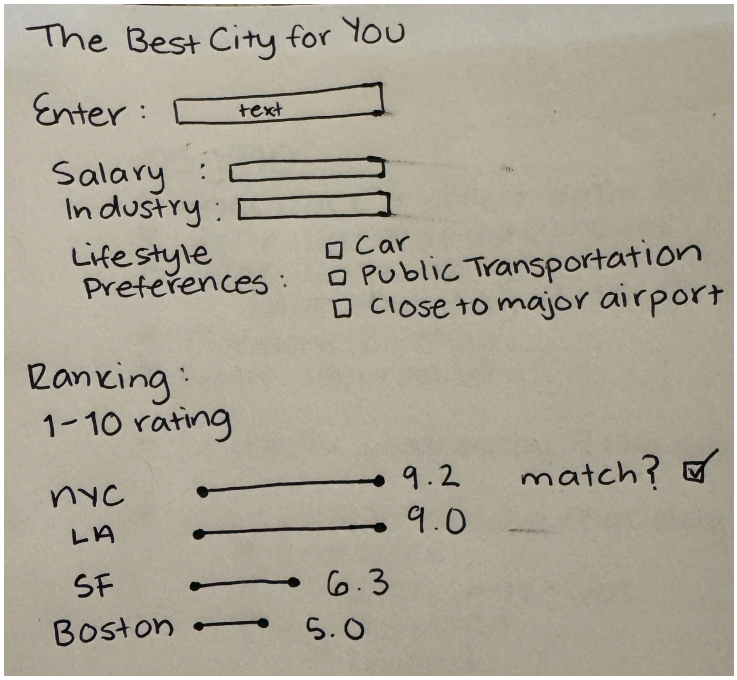
#6 How much of a recent graduate's monthly income would go towards rent in each city?
**Sketch ID 11 | Question ID 6, 18**



#4 Which U.S. states pay the most in taxes?
**Sketch ID 12 | Question ID 4**

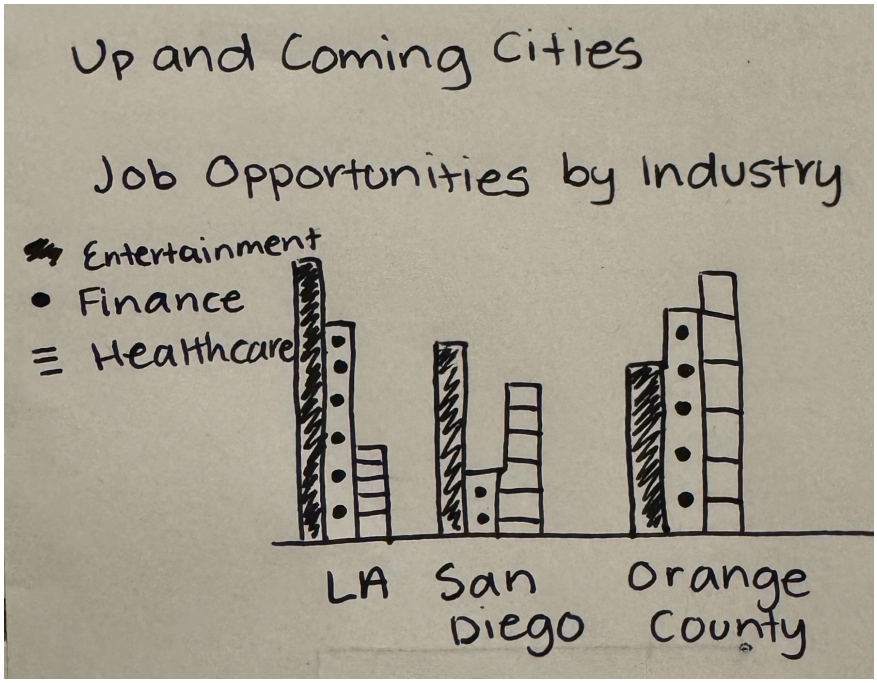



#17 How does the best city change depending on someone's salary, industry, and lifestyle priorities?
**Sketch ID 13 | Question ID 17**




#16 Which less-popular cities provide similar opportunities to major cities at a lower cost?

**Sketch ID 14 | Question ID 16**




#15 What expenses besides rent account for the largest differences in cost of living?

**Sketch ID 15 | Question ID 15**

# City Cost of Living Comparison

Enter: NYC Boston

| | | |
|---|---|---|
| Groceries | 2x less | 2x more |
| Gas | - | - |
| Entertainment | 6% less | 6% more |

## Milestone 4 | Decide

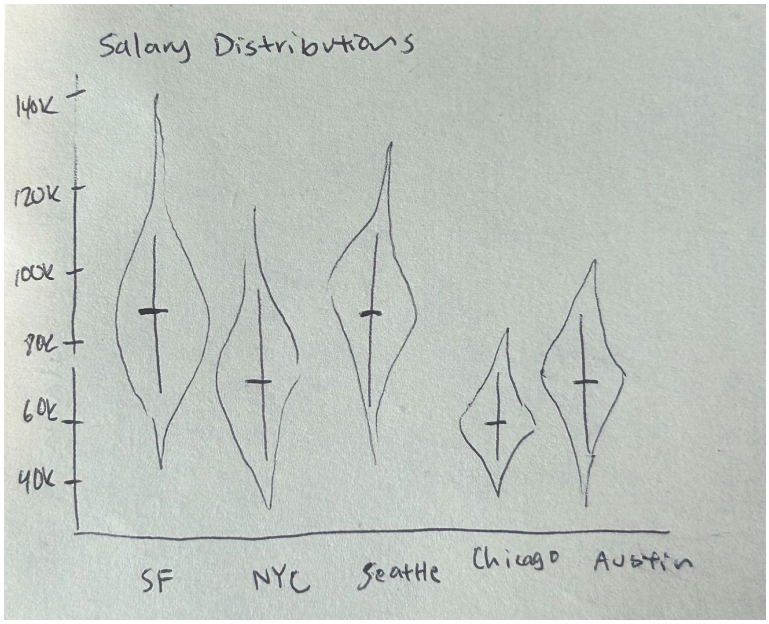
**Affinity Diagram and Voting**

| Sketch ID | Question ID | Author | Total Number of Votes |
|---|---|---|---|
| 1 | 5 | A.F. | 0 |
| 2, 11 | 18 | A.F., L.A. | 3 |
| 3 | 2 | A.F. | 1 |
| 4 | 4 | A.F. | 0 |
| 5 | 5 | A.F. | 3 |
| 6 | 7 | I.Q. | 3 |
| 7 | 8 | I.Q. | 1 |
| 8 | 9 | I.Q. | 0 |
| 9 | 11 | I.Q. | 1 |
| 10 | 12 | I.Q. | 0 |
| 12 | 4 | L.A. | 1 |
| 13 | 17 | L.A. | 2 |
| 14 | 16 | L.A. | 0 |
| 15 | 15 | L.A. | 0 |

After the voting process, we selected 4 sketches in total. Arranged logically, from most relevant to least relevant, they are:

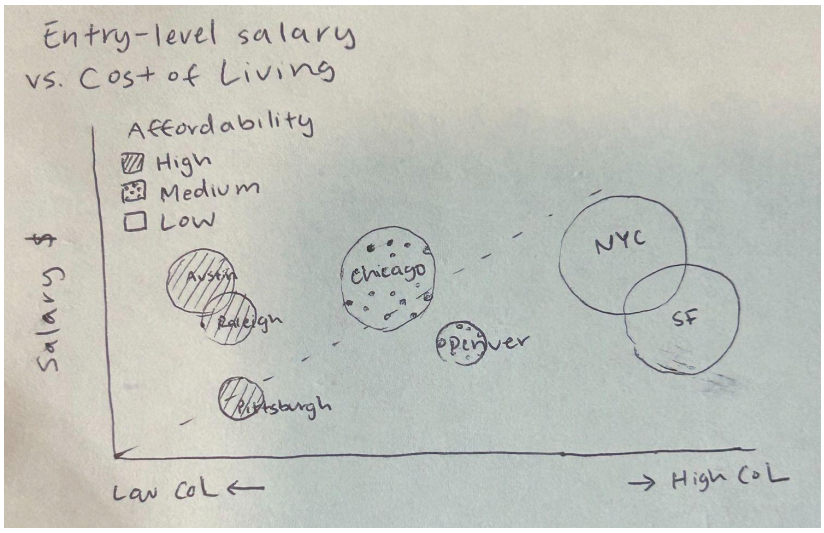
**Sketch 1**

**Q2: How can I compare cities by the average starting salary of their younger residents? This is stronger than showing only averages because it also displays the range and distribution of salaries.**



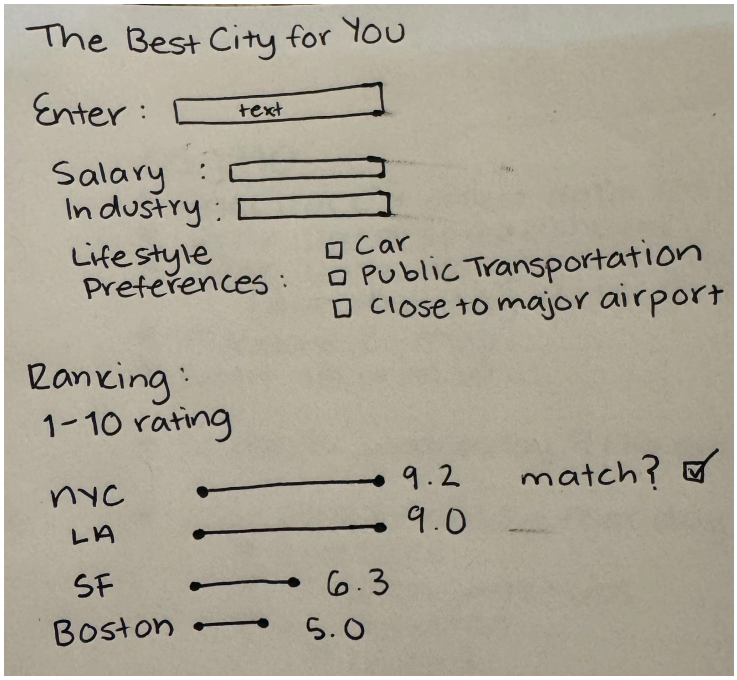

**Sketch 2**

**Q5:** Where does an entry-level salary go furthest after adjusting for cost of living? Also relates to Question 13 about career opportunities and affordability.



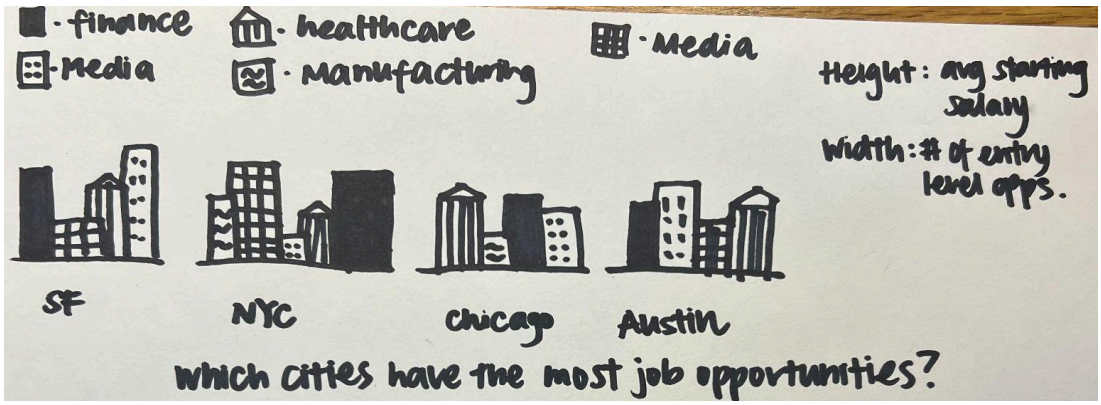

**Sketch 3**

**Q17:** How does the best city change depending on someone's salary, industry, and lifestyle priorities?




**Sketch 4**

**Q7:** Which cities have the most job opportunities for recent graduates?




**Reflection** - Add a one-paragraph explanation at the end summarizing your decisions and rationale for choosing the sketches you plan to implement.

We chose to implement the above 4 sketches, each receiving either 3 or 2 votes from the group. We approached the decision process by placing ourselves in the shoes of our audience, as upperclassmen nearing graduation but still deciding their career plans geographically. A lot of the factors driving relocation for post-graduation seem to be starting salary, commuting methods, and a growing population of peers. With these factors in mind, we selected visualizations that touch on those exact concerns, allowing the audience to gain relevant information in their post-graduation planning process. The third sketch selected takes a more exploratory than explanatory approach, in which the audience is able to interact with the visualization to receive information catered towards their specific situation. This is a slightly

more complex visualization that requires a model to rate cities based on the audience's choices, but we figured it's the type of tool that would actually be helpful towards our audience. Outside of this visualization, the remaining sketches use unique encoding techniques, such as area and hue, or graph styles, like the violin.

# Milestone 5 | Design Review

3-4 minute walkthrough
(a) your topic and the story you want to tell,

TOPIC: Post-graduation plans for early career professionals, especially regarding popular cities for relocation, salary, and housing. Our visualization tool will blend the explanatory and exploratory methods, allowing our target population to use it as part of their post-graduation planning.

The story: “Where can a recent graduate find work and make financial progress?”

We want to show how choosing a city affects both career opportunities and the life a graduate can afford there:

- Job opportunities: Which cities offer the most opportunities for recent graduates?
- Monthly savings: After rent and other living expenses, how much money could someone save?
- Savings goals: How does that monthly surplus translate into time needed to build an emergency fund or reach another goal?
- Personal fit: How do these tradeoffs change with industry, salary, and lifestyle priorities?

The takeaway: There’s no single best city for every graduate. **The visualization helps users identify where their career prospects, everyday spending, and savings goals align.**

(b) your data source and what it actually contains.

Every source reports data by metro area, and each metro has a standard government ID code (CBSA code). We match on that code: eg. Denver = 19740 in the jobs data, the Census data, and the price data

cities_README
cities.csv: data for 2025

- One combined dataset drawing from 6 public sources, spanning 40 major US metro areas.
    - Cost of living:
        - Economic Policy Institute Family Budget Calculator: what a single adult needs monthly for housing, food, transportation, healthcare, and taxes in each metro
    - Jobs & salaries:
        - Bureau of Labor Statistics wage data: employment counts and pay for every occupation in each metro. We kept the 132 occupations that typically require a bachelor's degree and no experience = "entry-level grad jobs"
    - People, rent & commutes:

        - Census American Community Survey: population aged 20–29, education levels, rent by apartment size, commute times, and how people get to work
    - Price index over time:
        - Bureau of Economic Analysis: how expensive each metro is relative to the U.S. average, and how that's changed
    - Unemployment:
        - Bureau of Labor Statistics, one rate per metro
    - Gas prices:
        - AAA state averages

Attributes:
city, state, region, cbsa_code, lat, lon, epi_annual_total, epi_effective_tax_rate, epi_monthly_housing, epi_monthly_food, epi_monthly_transportation, epi_monthly_healthcare, epi_monthly_other_necessities, epi_monthly_taxes, epi_monthly_total, epi_monthly_expenses_excl_taxes, total_jobs, median_salary_all_workers, entry_level_jobs, entry_salary_p10, entry_salary_p25, entry_salary_median, num_occupations, entry_jobs_per_1000, population, pop_age_20_29, share_age_20_29, mean_commute_min, pct_commute_transit, pct_commute_walk, pct_work_from_home, pct_commute_drive_alone, rent_studio, rent_1br, rent_2br, rent_3br, rent_per_person_2br_shared, rent_per_person_3br_shared, rpp_all_items, rpp_goods, rpp_housing, rpp_other_services, rpp_utilities, unemployment_rate, pct_bachelors_plus, median_household_income, median_gross_rent_all, gas_regular, est_monthly_take_home, est_monthly_savings, rent_share_of_take_home, months_to_save_10k, can_afford_studio_alone, salary_col_adjusted

city_trends.csv: the same 40 metros, 2008-2024. Answers longitudinal data questions
- Price level and housing costs per year (BEA)
- Per-capita income per year (BEA)
- Median household income & rent, 2021-2024 (Census ACS)

Attributes:
city, state, region, cbsa_code, year, rpp_all_items, rpp_housing, price_deflator, per_capita_income, median_household_income, median_gross_rent, rent_share_of_income

(c) the key questions you are answering,
- **Q2:** How can I compare cities by the average starting salary of their younger residents? This is stronger than showing only averages because it also displays the range and distribution of salaries.
- **Q5:** Where does an entry-level salary go furthest after adjusting for cost of living? Also relates to Question 13 about career opportunities and affordability.
- **Q17:** How does the best city change depending on someone’s salary, industry, and lifestyle priorities?
- **Q7:** Which cities have the most job opportunities for recent graduates?

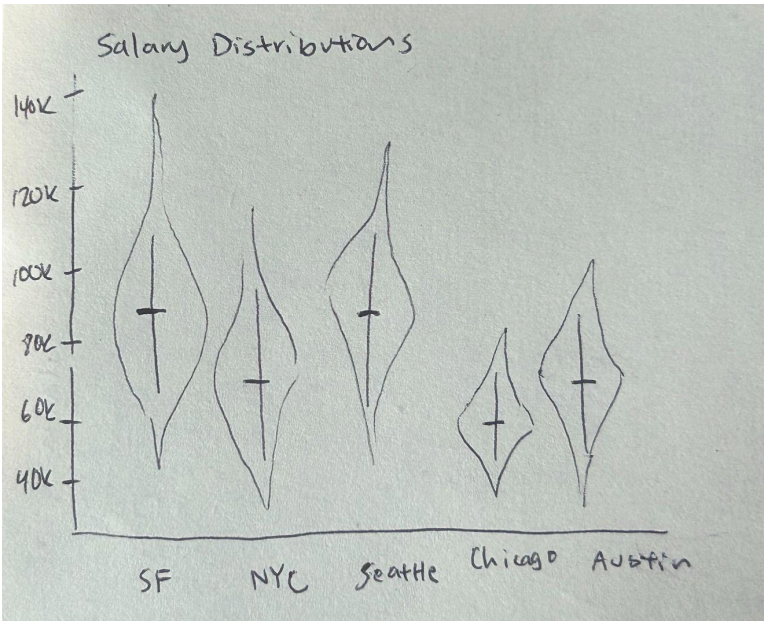
Why:

- We wanted to help college students compare cities for their first move after graduation by showing starting salary distributions, purchasing power, and job opportunities together. This gives a fuller picture of each city's economic prospects than average salaries alone.
- We chose these questions because salary ranges show the variation graduates might expect, cost-of-living adjustments reveal where those earnings go furthest, and job opportunities help ground the comparison in career prospects. Exploring how the best city changes with salary, industry, and lifestyle priorities also makes the visualization relevant to different graduates.

(d) the sketches you selected and why.

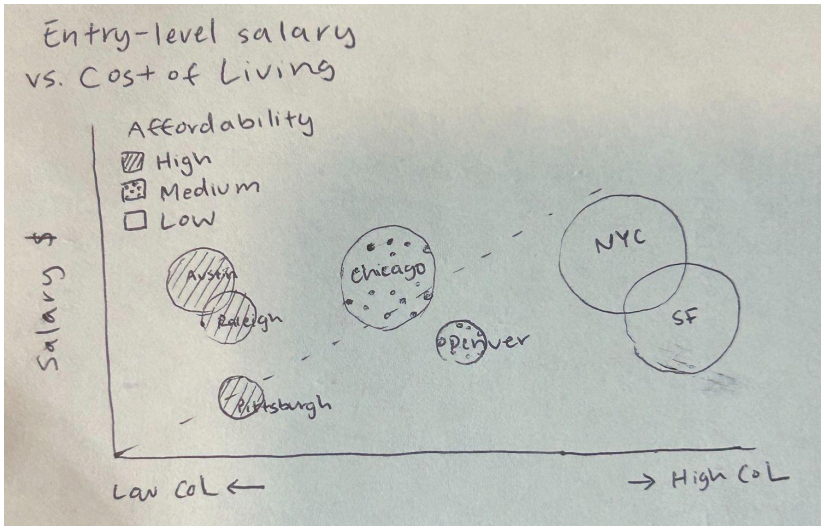
**Sketch 1**

**Q18:** How much could a recent graduate expect to save each month in different cities? Also answers Question 6 on rent and Question 15 on other expenses.



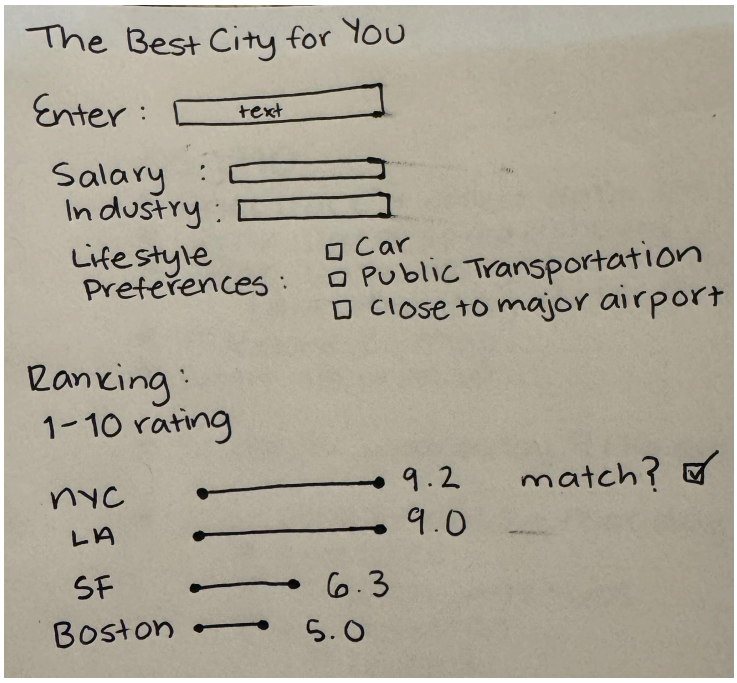

**Sketch 2**

**Q5:** How long would it take to reach a savings goal in each city?



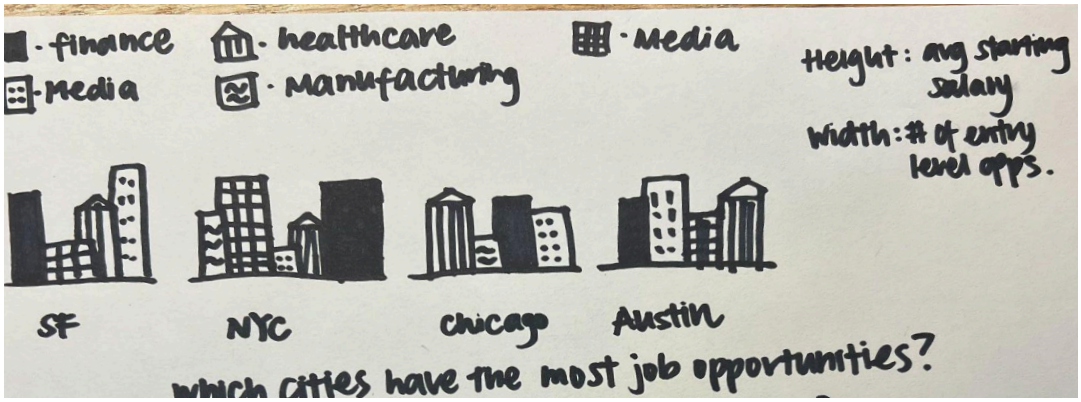

**Sketch 3**

**Q17:** How does the best city change depending on someone's salary, industry, and lifestyle priorities?




**Sketch 4**

**Q7:** Which cities have the most job opportunities for recent graduates?




**Reflection**

Placing ourselves in the shoes of our audience, as upperclassmen nearing graduation but still deciding their career plans geographically.

A lot of the factors driving relocation for post-graduation seem to be **starting salary, commuting methods, and a growing population of peers**. Sketch mapped to each concern area

- Sketches use unique encoding techniques, such as area and hue, or graph styles, like the violin.

The third sketch selected takes a more exploratory than explanatory approach, in which the audience is able to interact with the visualization to receive information catered towards their specific situation, MORE DYNAMIC.

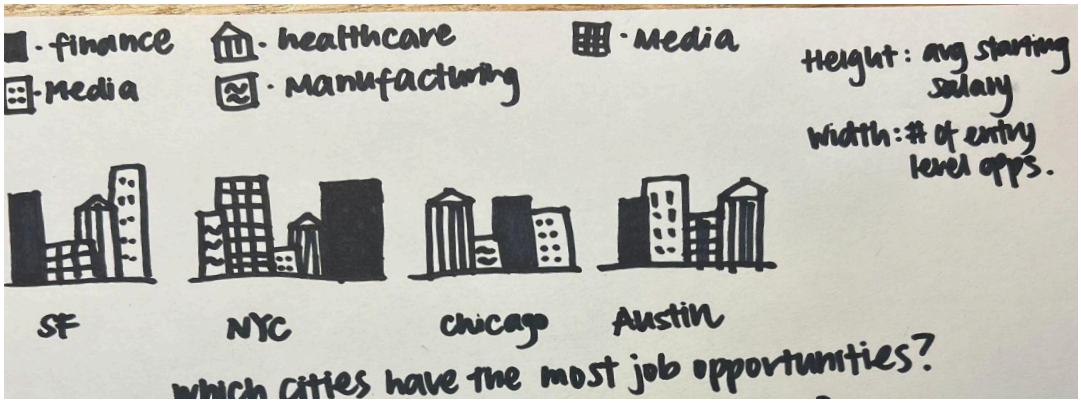
- INTERACTION LECTURE
    - Thinking about the visualization tool our audience of undergraduate students (upperclassmen level) would need:
    - Able to explore on their own an overview, then zoom in and see details on demand
    - Exploration-filter-elaborate

## Milestone 6 | Interactive Dashboards

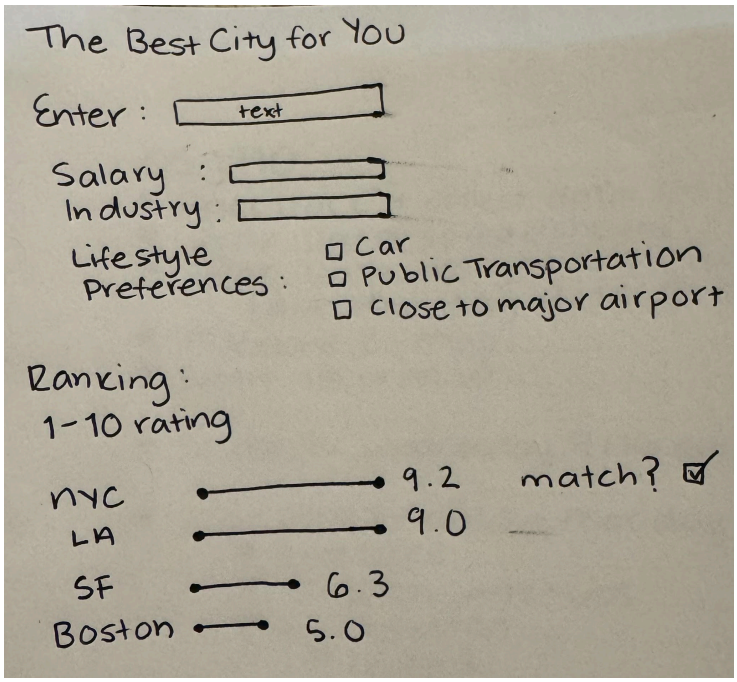
# 1. Selected Sketches and Rationale for Combining

The sketches we will be combining are:

**Sketch 4**
**Q7:** Which cities have the most job opportunities for recent graduates?




**Sketch 3**
**Q17:** How does the best city change depending on someone's salary, industry, and lifestyle priorities?

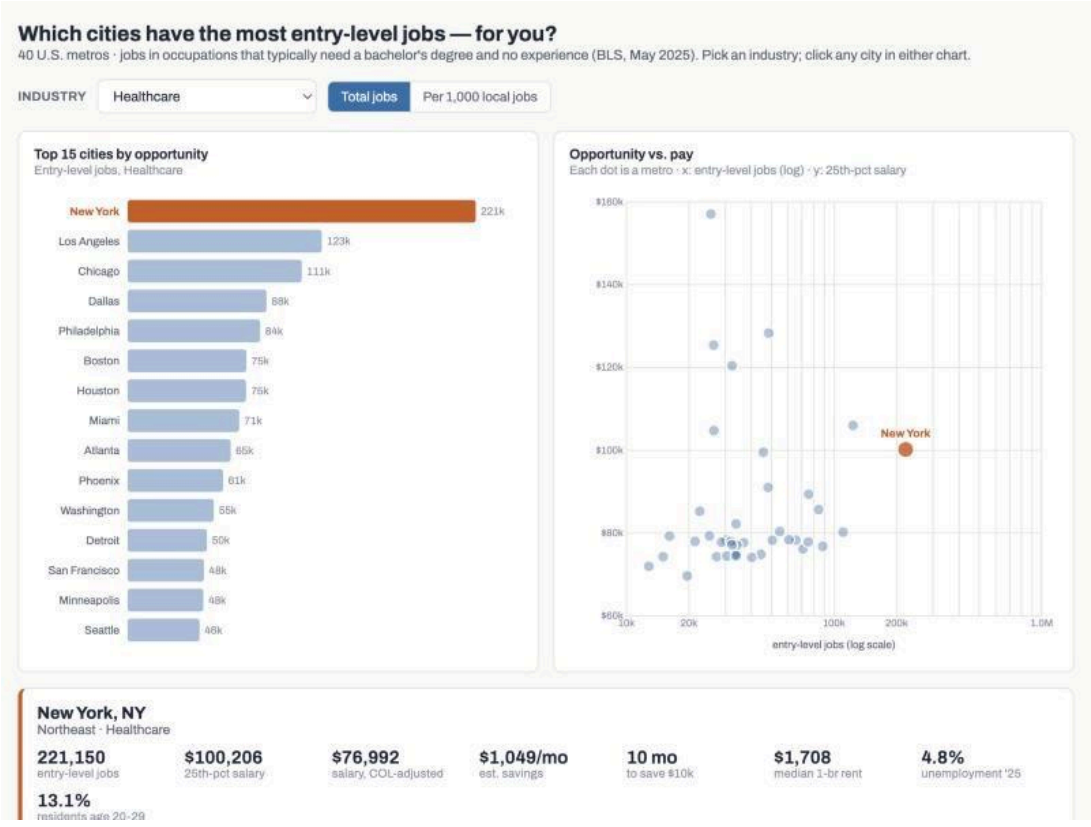



**Rationale:** Reflecting on our TF meeting with Yida, we want to pull some of the user interaction features from Sketch 3 and implement them in the view of Sketch 4. Since

we are really honing in on this visualization tool being both exploratory and explanatory for users, we want to present the industry data in a way that is both informing the user of the fields they can break into while also allowing them to explore specific industries if they already know where they're heading. So, in combining the elements of these sketches into a single view, we hope to open the tool for users to first receive an overview of the prominent industries across cities, then receive details on demand by selecting a specific industry they want to learn more about as an interaction feature. This combination is useful because it accounts for the varying intentions our users may have. While they are all in their early career, our audience certainly has a mix of people prepared to break into a field and others hoping to simply explore what's out there. With these sketches combined, the audience can answer: Which cities have the most job opportunities for recent graduates, especially for me?

## 2. Interactive Prototypes: Screenshots and Links

**Prototype 1:**




**Prototype 2:**

## Where are the jobs for new grads?

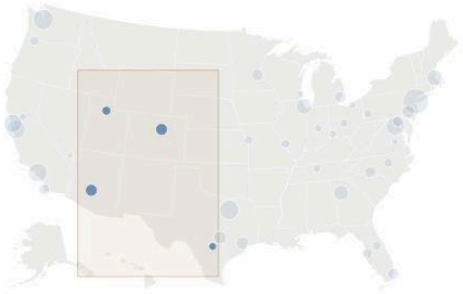
Bubble size = entry-level jobs (occupations needing a bachelor's degree, no experience · BLS May 2025). Pick an industry, choose a region — or drag a box on the map.

INDUSTRY: Computer & Math

All regions | Northeast | Midwest | South | West | Clear brush ×

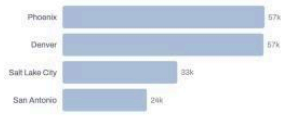
**U.S. metro map**

Drag to brush an area · click a bubble for details



**Top cities — brushed area (4)**

Computer & Math · entry-level jobs




Click a bubble or a bar to see a city's full profile.

## Where are the jobs for new grads?

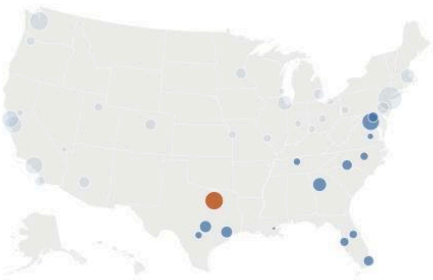
Bubble size = entry-level jobs (occupations needing a bachelor's degree, no experience · BLS May 2025). Pick an industry, choose a region — or drag a box on the map.

INDUSTRY: Computer & Math

All regions | Northeast | Midwest | South | West

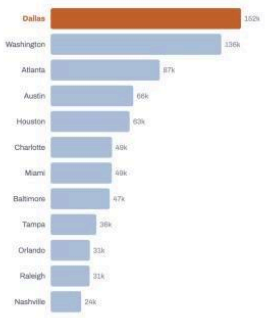
**U.S. metro map**

Drag to brush an area · click a bubble for details



**Top cities — South**

Computer & Math · entry-level jobs



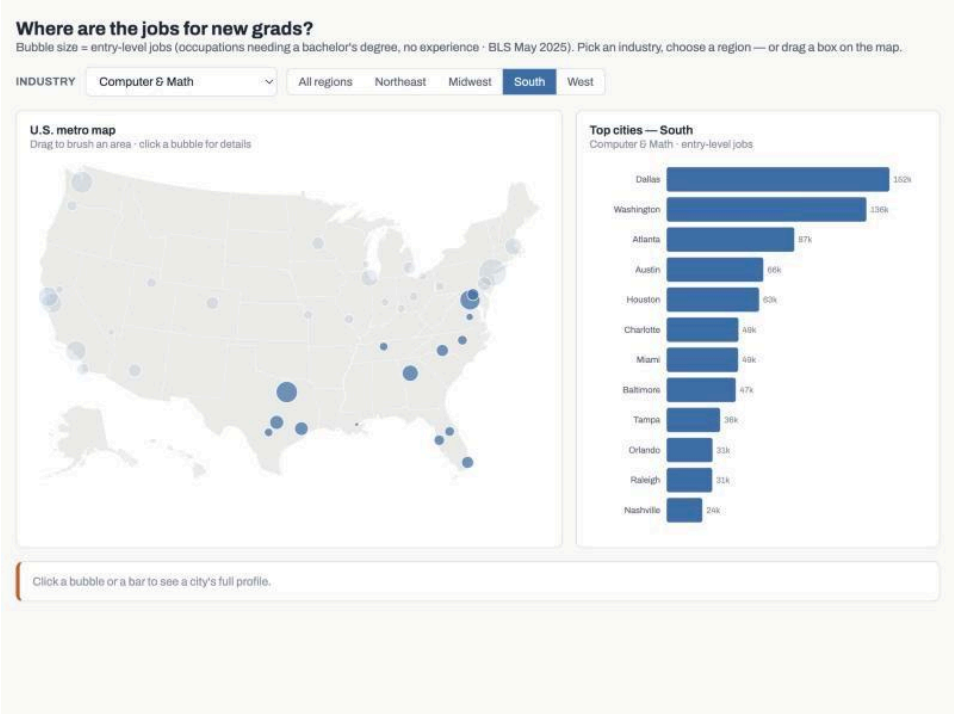

**Dallas, TX**

South · Computer & Math

| 151,780 | $94,692 | $70,978 | $2,644/mo | 4 mo | $1,473 | 3.9% |
|---|---|---|---|---|---|---|
| entry-level jobs | 25th-pct salary | salary, COL-adjusted | est. savings | to save $10k | median 1-br rent | unemployment '25 |

**14.1%**

residents age 20-29




## 3. Interactive Prototypes: Descriptions

**Prototype 1:**

- Industry Selection (Global Filter / Dropdown): Users can select or search for a specific industry (or clear the selection to view the default aggregate overview). When an industry is selected, a shared filter is applied across all secondary charts in the view. The visualization will change to only present information from that targeted industry.
- City Selection: Users can select from a list of cities and the secondary charts in the view will highlight that city in a more prominent color. The information card at the bottom will display details about that city's overall workforce, including its entry level jobs, average salary after adjusting for cost of living, and young residents.

**Prototype 2:**

- Brushing a Region: Brushing a geographical region automatically highlights the top cities found within that brushed region. When an industry is selected, a shared filter is applied across all views.
- Industry Selection: Users can select a specific industry (or clear the selection to view the default aggregate overview).

- Selecting a City: Selecting a specific city automatically highlights corresponding data points across the industry distribution breakdown and detailed metric panels. The information card at the bottom will display details about that city's overall workforce, including its entry level jobs, average salary after adjusting for cost of living, and young residents.

- Differs from Prototype 1 in that the implementation of a map of the U.S. allows for brushing and selecting data points for cities.

## 4. Design Comparison

Prototype 1 is better at the tradeoff question: the scatter makes it obvious that NYC has the most healthcare jobs, but has mid-level pay. Solely looking at ranking would hide this. Prototype 2 is better at the geography question: the user is able to instantly pick which area to look at, which is tedious in a sorted list. A weakness of the view includes that the bubble area is hard to precisely compare, and so we incorporate the linked bar chart. A possible next step could be merging these two prototypes together, so that we have a map for orientation, a scatter for salaries, and a shared filter row.

## 5. Claude Explanation

We gave Claude our cleaned dataset and our interaction plan and it built both prototypes in D3, embedding the real data. We verified the interaction works by testing it and ensuring all the filters and brushing worked. We also changed our question from which cities have the most job opportunities for recent graduates, especially for me, to where are the jobs for new grads, from prototype 1 to 2, as a result of the map adding a more visual geographical aspect.

## Milestone 7 | Storyboards

# Team's Data Insights

**Anna:**

- The Midwest is where a starting salary goes furthest. Nine of its ten metros save more per month than the median city ($1,102), and the regional median is $1,430 against $944 in the West, $782 in the Northeast and $582 in the South. Five of the top ten savers nationally are Midwestern, led by Detroit ($1,968), Cleveland ($1,634) and Cincinnati ($1,589). The reason is cost, not pay. Detroit's starting salary is $73.7k, barely above the median city's $70.1k, but rent takes only 20% of take-home pay there, against 32–35% in the other regions. Chicago is the exception at $889 a month with rent at 31% of take-home.
- Higher pay doesn't predict higher savings. Starting salary correlates 0.82 with the cost of a modest budget but only 0.28 with monthly savings. New York has the fourth-highest starting salary ($86.7k) but the lowest savings of all 40 cities ($211 a month), with rent at 45% of take-home. Boston ranks fifth on pay and 38th on savings ($395). Detroit pays about $13k less than New York but leaves a graduate roughly $1,750 more each month.
- Rent is outrunning income fastest in the Sun Belt, not in the expensive coastal metros. From 2021 to 2024, median rent grew faster than median household income in 36 of 39 cities. Tampa (+38% rent vs. +24% income), Miami (+37% vs. +26%) and Phoenix (+31% vs. +19%) saw the biggest jumps in rent share of income, up 2.3–2.7 points each. The only cities where rent share fell were San Francisco, New York and San Jose (down 0.5–0.8 points). That's a concern for Miami and Tampa, which already save only $298 and $553 a month.

**Lorena:**

- San Jose has the highest concentration of grad-level work of any city in the dataset, at 292.8 grad jobs per 1,000 jobs against a median of 180. What stands out is how much of its economy is grad-level work. Its estimated starting salary is $128,515, nearly double the median city's $70,140. That leaves it first on pay even after adjusting for cost of living ($116,384).
- Western metros have the highest median starting salary of any region ($81k vs. $68–79k elsewhere), and four of the five densest grad job markets are here: San Jose, Seattle, Denver and San Francisco. That advantage is uneven, though. The West also has the highest median unemployment (4.6%), and Las Vegas has the fewest grad jobs per 1,000 jobs (112.2) and the highest unemployment (5.6%) of all 40 cities.
- Southern metros are the easiest places to get hired: they hold 7 of the 10 lowest unemployment rates, led by Nashville (3.0%) and Raleigh (3.2%). Their pay is among the lowest, though, so six Southern cities land in the bottom 10 for monthly savings,

including Atlanta, Miami and Nashville at about $300 a month. Texas metros like Houston ($1,455) and San Antonio ($1,363) save the most in the region.

**Ivy:**

- Across our 40 metros, the correlation between entry-level salary and estimated monthly savings is only r = 0.28. New York pays the 4th-highest entry salary ($86.7k) but ranks dead last (40th) in monthly savings ($211/mo). Boston falls from 5th to 35th. Meanwhile Cleveland and Cincinnati rank 28th in salary but 3rd/4th in savings (~$1,600/mo). The one city that beats this is San Jose: tech salaries are so high ($128.5k) that it leads in savings despite the highest rents.
- Using the 2008-2024 trends file: Tampa (+22), Phoenix (+20), and Orlando (+18.5) gained the most housing-price-level points relative to the US average since 2015, while San Jose (−24), Washington (−20), and New York (−16) fell the most. The same shows up short-term: rent as a share of income rose most from 2021-2024 in Phoenix, Miami, Tampa, and Richmond, and fell in SF, NYC, and San Jose. The affordability gap is converging from both ends, contradicting the traditional view that moving south is cheap.
- The median within-city spread in monthly savings across career fields is $3,468 which is bigger than the entire spread across all 40 cities ($2,067). In Boston, a CS grad saves ~$1,966/mo while an arts/media grad runs ~$1,254/mo in the red on the same “modest but adequate” budget. So “what’s the best city?” is unanswerable without asking the user’s industry first.

# Central Message - So what?

**Central Message:**

A higher starting salary doesn’t mean you’ll keep more of it: New York pays the fourth-highest in our data ($86.7k) but saves the least ($211 a month), while Detroit pays about $13k less and saves roughly $1,750 more, so graduates should compare cities by what they keep, not just what they earn.

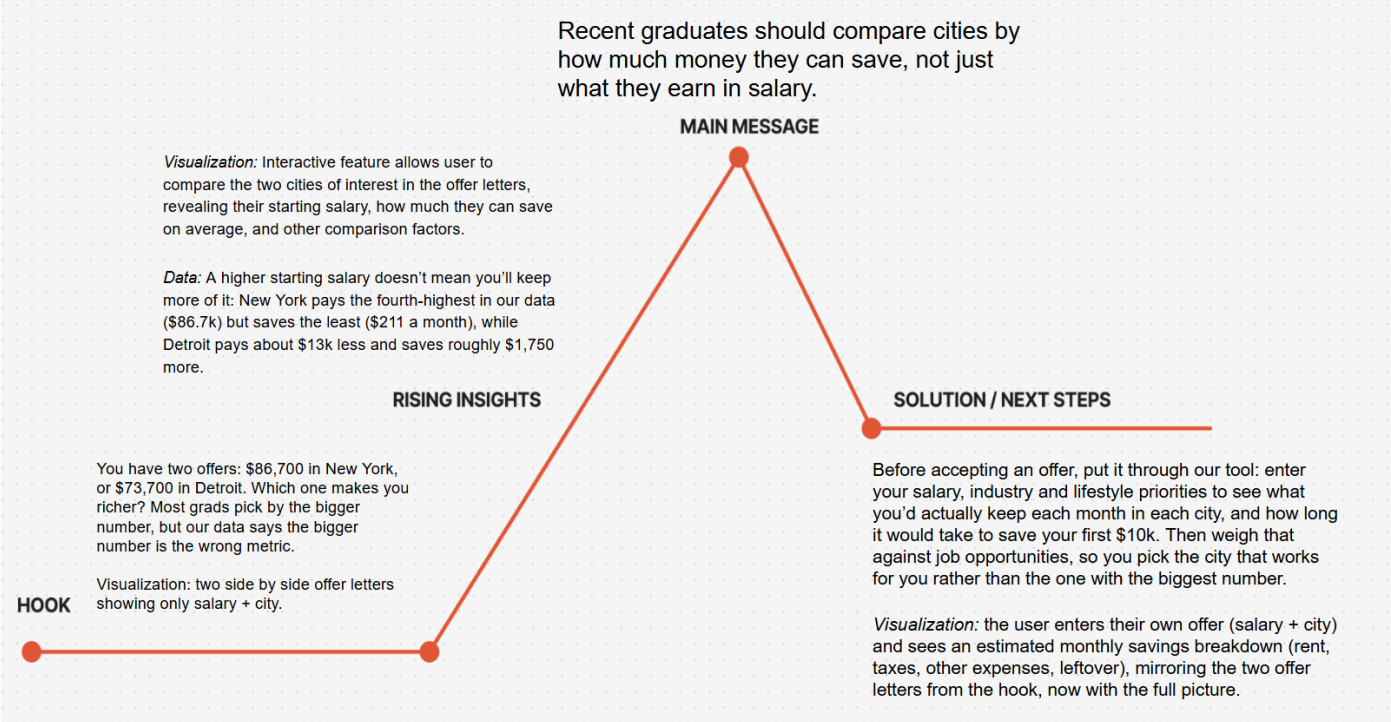
**Explanation:**

We chose this insight because it directly answers our story question, “Where can a recent graduate find work and make financial progress?” It is also the clearest evidence for why our tool combines job opportunity, cost-adjusted pay and personalized filters instead of ranking cities by salary alone. Across the 40 metros, starting salary correlates 0.82 with the cost of a modest budget but only 0.28 with monthly savings, so the headline salary a graduate sees is a poor guide to their financial progress. The insight is also relevant to our audience, since upperclassmen comparing offers will most likely compare salaries first. The Midwest savings pattern and the Sun Belt rent trend support this message but serve as secondary points. One

caveat to note: savings are estimated at the 25th-percentile salary using EPI's budget, which assumes car ownership everywhere and so likely understates savings in transit cities like New York.

# Data Storyboard

**Screenshot and Link:**
[Data Storyboard]



**Milestone 8 | Prototype I**

# Team Member Documentation

We met as a team to draft, design, and implement our prototype. The team members that actively worked on the prototype are Lorena Alvarez, Anna Feng, and Ivy Qin.

# Complete Data Pipeline

The data pipeline was created during Milestone 5 as part of our preparation for the Design Review meeting with our TF Yida Chen. We merged our real datasets according to the process outlined below:

Every source reports data by metro area, and each metro has a standard government ID code (CBSA code). We match on that code: eg. Denver = 19740 in the jobs data, the Census data, and the price data

cities_README
cities.csv: data for 2025

One combined dataset drawing from 6 public sources, spanning 40 major US metro areas.
Cost of living:
Economic Policy Institute Family Budget Calculator: what a single adult needs monthly for housing, food, transportation, healthcare, and taxes in each metro
Jobs & salaries:
Bureau of Labor Statistics wage data: employment counts and pay for every occupation in each metro. We kept the 132 occupations that typically require a bachelor's degree and no experience = "entry-level grad jobs"
People, rent & commutes:
Census American Community Survey: population aged 20–29, education levels, rent by apartment size, commute times, and how people get to work
Price index over time:
Bureau of Economic Analysis: how expensive each metro is relative to the U.S. average, and how that's changed
Unemployment:
Bureau of Labor Statistics, one rate per metro
Gas prices:
AAA state averages

Attributes:
city, state, region, cbsa_code, lat, lon, epi_annual_total, epi_effective_tax_rate, epi_monthly_housing, epi_monthly_food, epi_monthly_transportation, epi_monthly_healthcare, epi_monthly_other_necessities, epi_monthly_taxes, epi_monthly_total, epi_monthly_expenses_excl_taxes, total_jobs, median_salary_all_workers, entry_level_jobs,

entry_salary_p10, entry_salary_p25, entry_salary_median, num_occupations, entry_jobs_per_1000, population, pop_age_20_29, share_age_20_29, mean_commute_min, pct_commute_transit, pct_commute_walk, pct_work_from_home, pct_commute_drive_alone, rent_studio, rent_1br, rent_2br, rent_3br, rent_per_person_2br_shared, rent_per_person_3br_shared, rpp_all_items, rpp_goods, rpp_housing, rpp_other_services, rpp_utilities, unemployment_rate, pct_bachelors_plus, median_household_income, median_gross_rent_all, gas_regular, est_monthly_take_home, est_monthly_savings, rent_share_of_take_home, months_to_save_10k, can_afford_studio_alone, salary_col_adjusted

city_trends.csv: the same 40 metros, 2008-2024. Answers longitudinal data questions
Price level and housing costs per year (BEA)
Per-capita income per year (BEA)
Median household income & rent, 2021-2024 (Census ACS)

Attributes:
city, state, region, cbsa_code, year, rpp_all_items, rpp_housing, price_deflator, per_capita_income, median_household_income, median_gross_rent, rent_share_of_income

# Two Functional D3 Visualizations

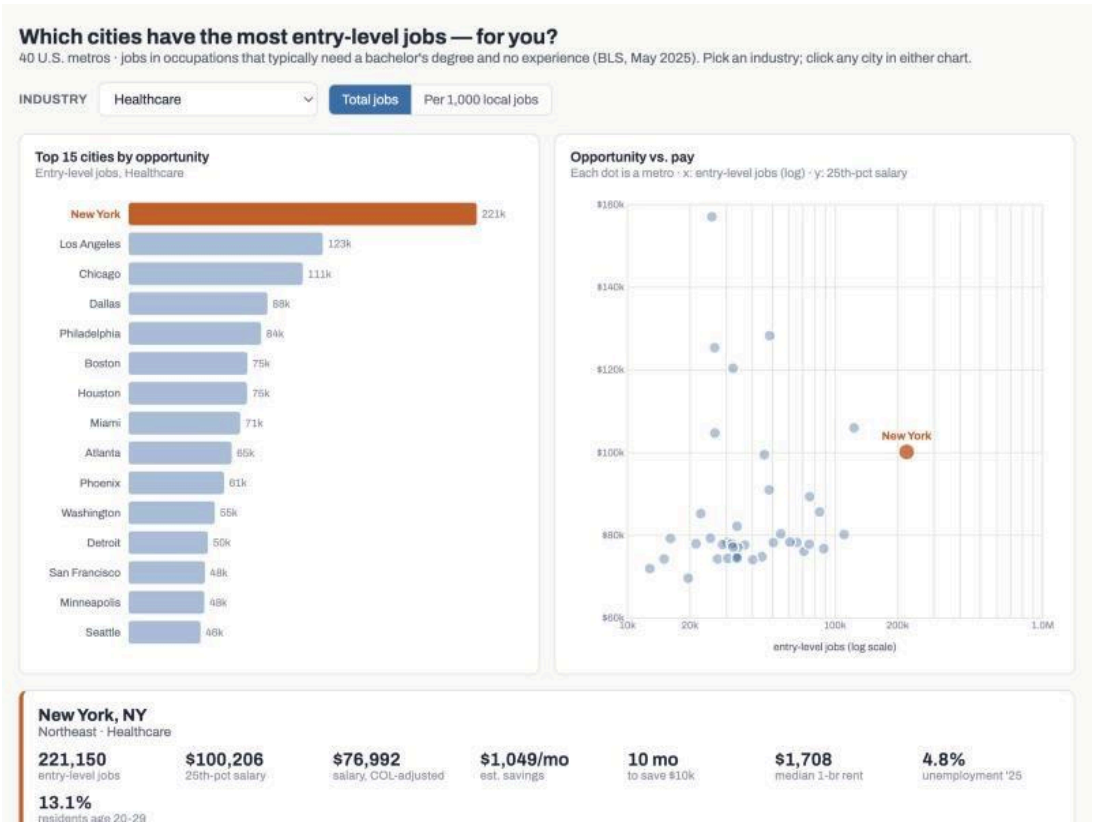
The two functional D3 visualizations we are choosing to highlight are the same mini-dashboards we previously created for MS 6. These fit our data story because:

- Prototype 1 contains an “Opportunity VS Pay” scatter plot and highlights a specific city. This works well for the “Rising Insights” of our data story because the user already selected two cities/salaries to compare from the hook, so they receive more information to aid their decision on where to relocate.

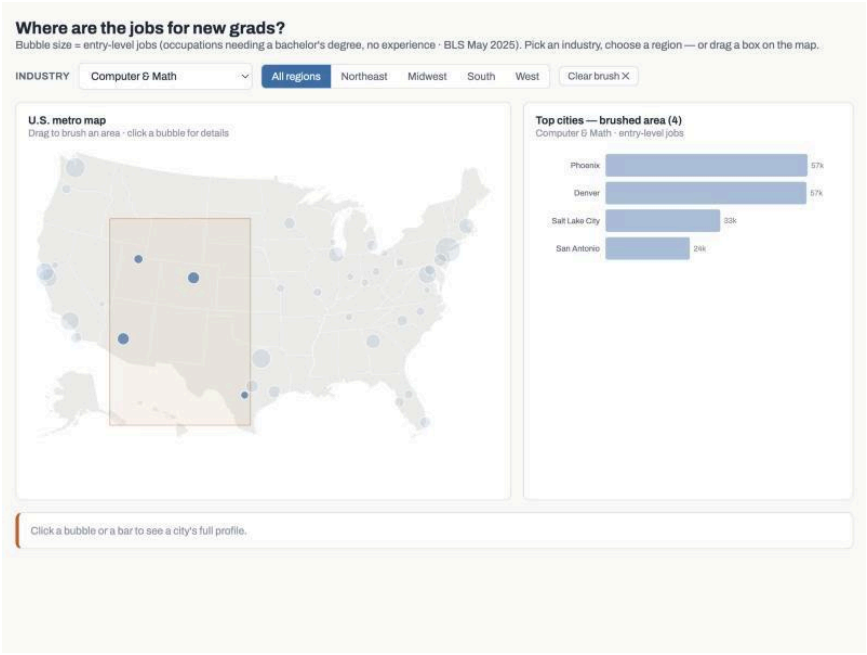
- Prototype 2 offers a lot of viewer interaction, which is in line with our data story’s “Next Steps”. After reading our story’s message, the user should enter their own offer (salary + city) and see an estimated monthly savings breakdown (rent, taxes, other expenses, leftover), mirroring the two offer letters from the hook, now with the full picture.

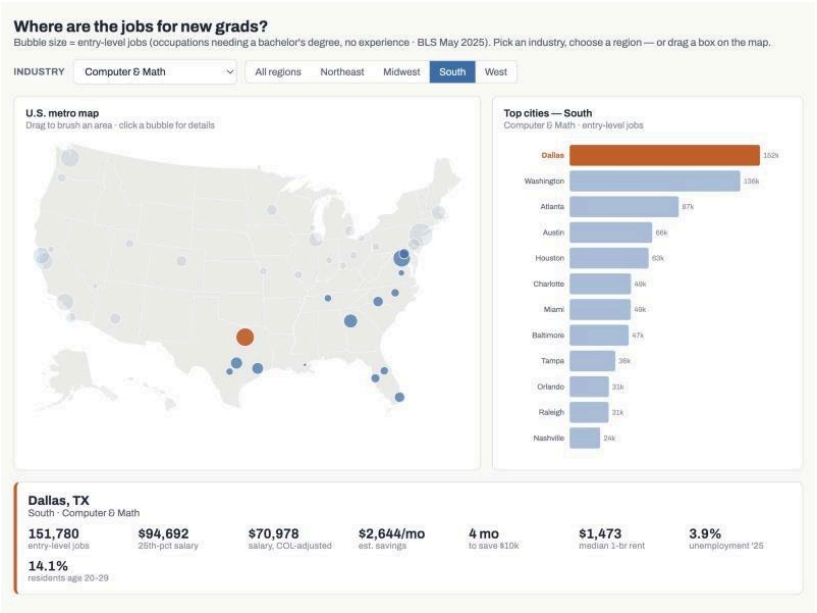
Their screenshots are found below:

**Prototype 1:**


Which cities have the most entry-level jobs — for you?
40 U.S. metros · jobs in occupations that typically need a bachelor's degree and no experience (BLS, May 2025). Pick an industry; click any city in either chart.
INDUSTRY
Healthcare
Total jobs
Per 1,000 local jobs
Top 15 cities by opportunity
Entry-level jobs, Healthcare
New York 221k
Los Angeles 123k
Chicago 111k
Dallas 88k
Philadelphia 84k
Boston 75k
Houston 76k
Miami 71k
Atlanta 65k
Phoenix 61k
Washington 55k
Detroit 50k
San Francisco 48k
Minneapolis 48k
Seattle 46k
Opportunity vs. pay
Each dot is a metro · x: entry-level jobs (log) · y: 25th-pct salary
$160k
$140k
$120k
$100k
$80k
$60k
New York
10k
20k
100k
200k
1.0M
entry-level jobs (log scale)
New York, NY
Northeast · Healthcare
221,150
entry-level jobs
$100,206
25th-pct salary
$76,992
salary, COL-adjusted
$1,049/mo
est. savings
10 mo
to save $10k
$1,708
median 1-br rent
4.8%
unemployment '25
13.1%

## Prototype 2:


Where are the jobs for new grads?
Bubble size = entry-level jobs (occupations needing a bachelor's degree, no experience · BLS May 2025). Pick an industry, choose a region — or drag a box on the map.
INDUSTRY
Computer & Math
All regions
Northeast
Midwest
South
West
Clear brush ×
U.S. metro map
Drag to brush an area · click a bubble for details
Top cities — brushed area (4)
Computer & Math · entry-level jobs
Phoenix 67k
Denver 67k
Salt Lake City 33k
San Antonio 24k
Click a bubble or a bar to see a city's full profile.




# Additional Visualization Drafts

Updated designs for 2-3 more visualizations in your process book that you haven't implemented yet.

The additional visualization drafts we are choosing to highlight are:

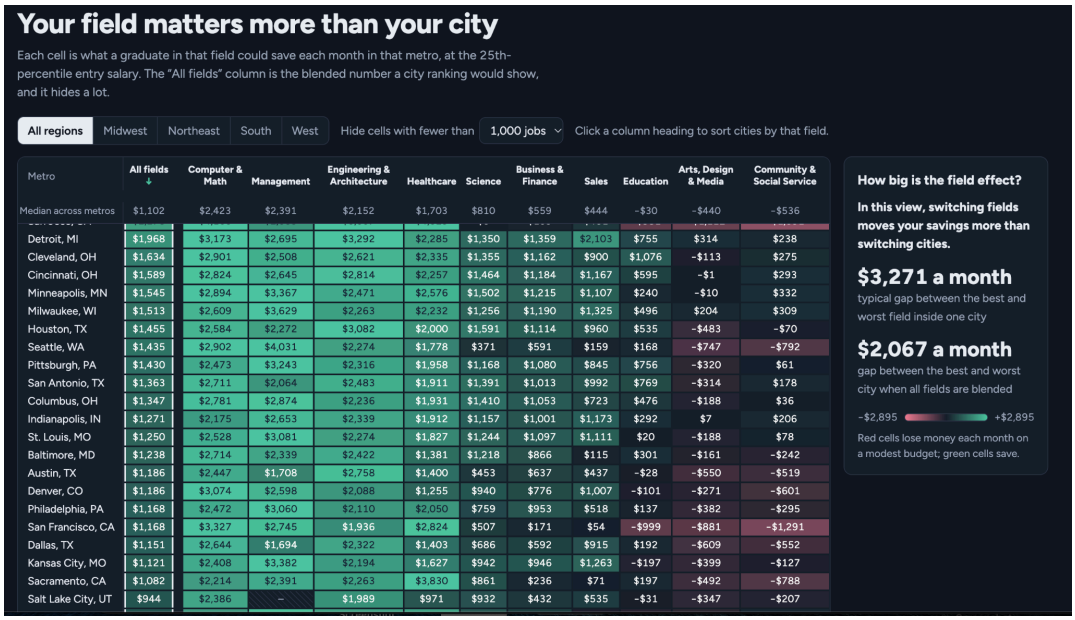
- Field Heatmap of monthly savings by city and career field.
- A savings-goal timeline showing how many months a typical entry-level salary needs to save a $10,000 emergency fund in each metro, with the viewer able to switch career field, filter by region, and change the goal amount.
- A revised version of our use of the U.S. map. The map view is static but the viewer can interact with dropdown / select features so that the map highlights different cities.

These fit our data story because

- It fits our data story because it shows that what a graduate keeps depends on their field as much as their city: the typical gap between the best and worst field within one city is about $3.3K a month, compared with about $2.1K between the best and worst city. This supports our main message that graduates should compare cities by what they keep, and it explains why our tool asks for industry before recommending a city.
- It fits our data story because it turns “what you keep” into time: on the default view Detroit reaches $10,000 in about 5 months while New York, which pays about $13K more, takes 47 months, and switching the field shows that on an arts or media salary most metros never get there at all.
- This map will serve to ground our viewer in the scenario that they are a new grad about to relocate for their first job. This is the hook to our story where the map highlights the two cities our viewer has been offered a job in.

Their screenshots are found below:

Additional Visualization #1



## Your field matters more than your city

Each cell is what a graduate in that field could save each month in that metro, at the 25th-percentile entry salary. The "All fields" column is the blended number a city ranking would show, and it hides a lot.

All regions | Midwest | Northeast | South | West — Hide cells with fewer than 1,000 jobs — Click a column heading to sort cities by that field.

| Metro | All fields ↓ | Computer & Math | Management | Engineering & Architecture | Healthcare | Science | Business & Finance | Sales | Education | Arts, Design & Media | Community & Social Service |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Median across metros | $1,102 | $2,423 | $2,391 | $2,152 | $1,703 | $810 | $559 | $444 | -$30 | -$440 | -$536 |
| Detroit, MI | $1,968 | $3,173 | $2,695 | $3,292 | $2,285 | $1,350 | $1,359 | $2,103 | $755 | $314 | $238 |
| Cleveland, OH | $1,634 | $2,901 | $2,508 | $2,621 | $2,335 | $1,355 | $1,162 | $900 | $1,076 | -$113 | $275 |
| Cincinnati, OH | $1,589 | $2,824 | $2,645 | $2,814 | $2,257 | $1,464 | $1,184 | $1,167 | $595 | -$1 | $293 |
| Minneapolis, MN | $1,545 | $2,894 | $3,367 | $2,471 | $2,576 | $1,502 | $1,215 | $1,107 | $240 | -$10 | $332 |
| Milwaukee, WI | $1,513 | $2,609 | $3,629 | $2,263 | $2,232 | $1,256 | $1,190 | $1,325 | $496 | $204 | $309 |
| Houston, TX | $1,455 | $2,584 | $2,272 | $3,082 | $2,000 | $1,591 | $1,114 | $960 | $535 | -$483 | -$70 |
| Seattle, WA | $1,435 | $2,902 | $4,031 | $2,274 | $1,778 | $371 | $591 | $159 | $168 | -$747 | -$792 |
| Pittsburgh, PA | $1,430 | $2,473 | $3,243 | $2,316 | $1,958 | $1,168 | $1,080 | $845 | $756 | -$320 | $61 |
| San Antonio, TX | $1,363 | $2,711 | $2,064 | $2,483 | $1,911 | $1,391 | $1,013 | $992 | $769 | -$314 | $178 |
| Columbus, OH | $1,347 | $2,781 | $2,874 | $2,236 | $1,931 | $1,410 | $1,053 | $723 | $476 | -$188 | $36 |
| Indianapolis, IN | $1,271 | $2,175 | $2,653 | $2,339 | $1,912 | $1,157 | $1,001 | $1,173 | $292 | $7 | $206 |
| St. Louis, MO | $1,250 | $2,528 | $3,081 | $2,274 | $1,827 | $1,244 | $1,097 | $1,111 | $20 | -$188 | $78 |
| Baltimore, MD | $1,238 | $2,714 | $2,339 | $2,422 | $1,381 | $1,218 | $866 | $115 | $301 | -$161 | -$242 |
| Austin, TX | $1,186 | $2,447 | $1,708 | $2,758 | $1,400 | $453 | $637 | $437 | -$28 | -$550 | -$519 |
| Denver, CO | $1,186 | $3,074 | $2,598 | $2,088 | $1,255 | $940 | $776 | $1,007 | -$101 | -$271 | -$601 |
| Philadelphia, PA | $1,168 | $2,472 | $3,060 | $2,110 | $2,050 | $759 | $953 | $518 | $137 | -$382 | -$295 |
| San Francisco, CA | $1,168 | $3,327 | $2,745 | $1,936 | $2,824 | $507 | $171 | $54 | -$999 | -$881 | -$1,291 |
| Dallas, TX | $1,151 | $2,644 | $1,694 | $2,322 | $1,403 | $686 | $592 | $915 | $192 | -$609 | -$552 |
| Kansas City, MO | $1,121 | $2,408 | $3,382 | $2,194 | $1,627 | $942 | $946 | $1,263 | -$197 | -$399 | -$127 |
| Sacramento, CA | $1,082 | $2,214 | $2,391 | $2,263 | $3,830 | $861 | $236 | $71 | $197 | -$492 | -$788 |
| Salt Lake City, UT | $944 | $2,386 | – | $1,989 | $971 | $932 | $432 | $535 | -$31 | -$347 | -$207 |

**How big is the field effect?**

**In this view, switching fields moves your savings more than switching cities.**

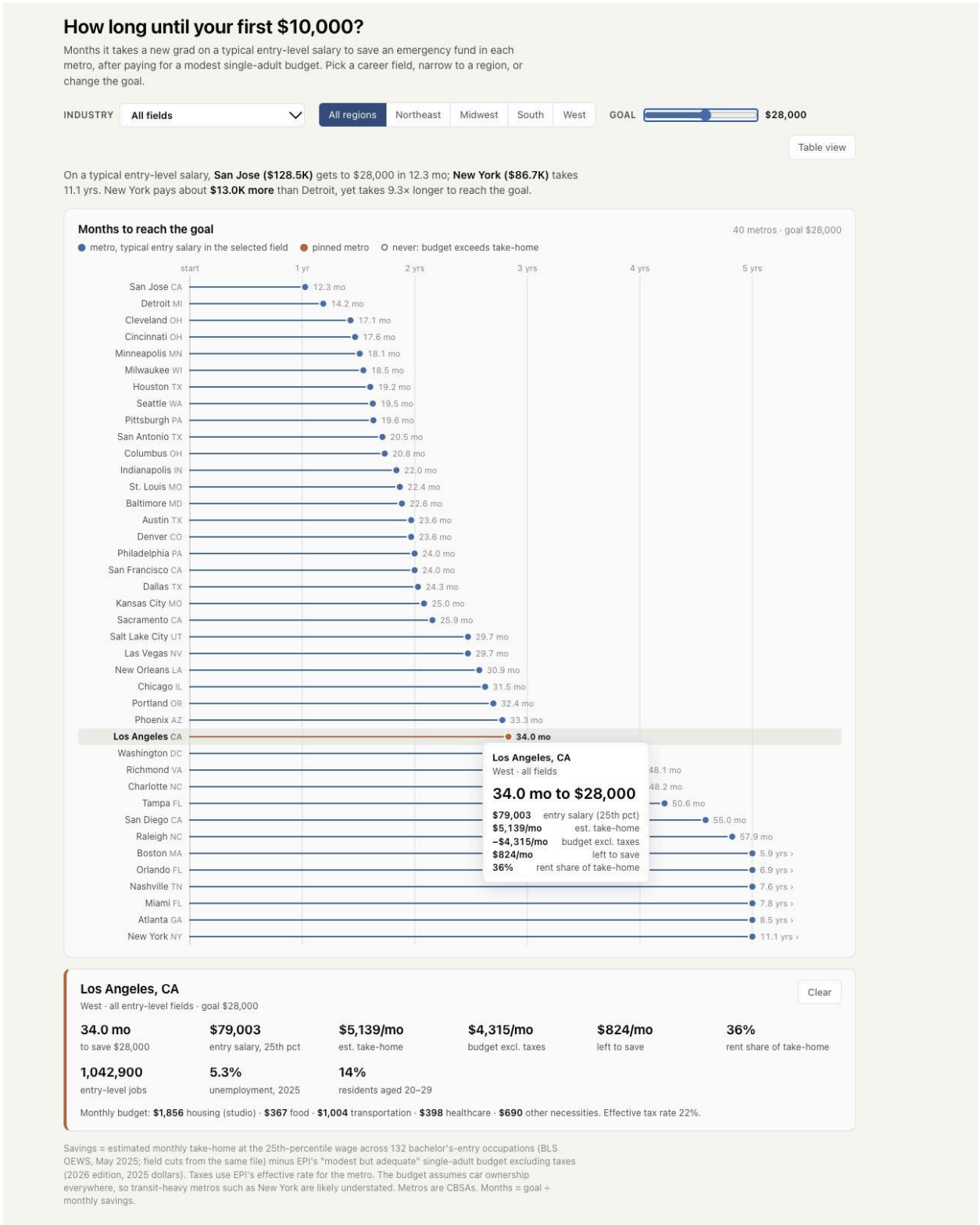
**$3,271 a month**
typical gap between the best and worst field inside one city

**$2,067 a month**
gap between the best and worst city when all fields are blended

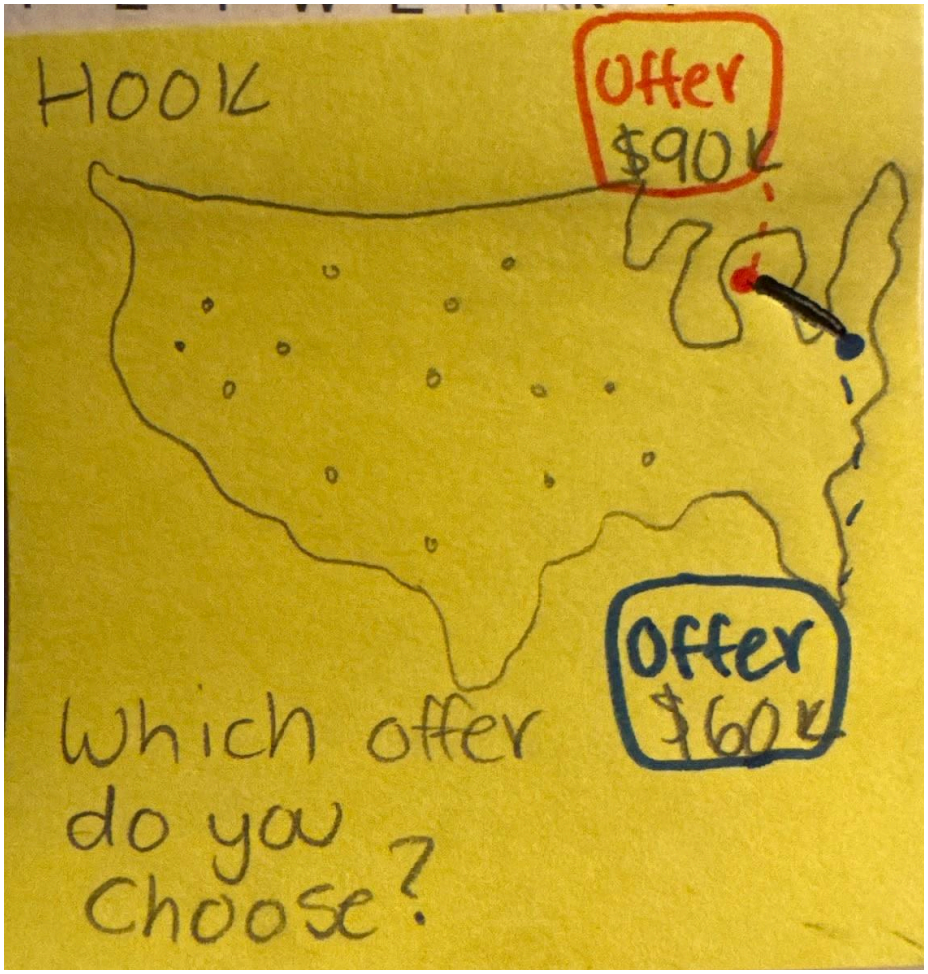
-$2,895 +$2,895

Red cells lose money each month on a modest budget; green cells save.

Additional Visualization #2




Additional Visualization (Sketch) #3




## Website Structure

Website structure: Rough webpage design and layout implemented (placeholders for visualizations, text, and images are acceptable)

https://claude.ai/artifact/6Pm8QaZxsRLrfyH7EywvVB?sk=hhJJZsLyYNyJNGj2FhrLTw

## Storytelling

The current structure of the website is designed after our storyboard implemented in MS 7. We want to continue with the message that there are other factors outside of starting salary that should be considered when choosing where to relocate for work.

## Novel View Design

One of the visualization drafts we chose was to revise the purpose of a map in our data story. Initially, the map was simply part of the interaction visualization in which the user selected a

specific city to learn more about. Considering our data story from MS 7, we chose to incorporate a map as part of our hook. The viewer is presented to a default overview of the map in which two cities are highlighted. The viewer has received offer letters from these two cities and must decide where to relocate based on the starting salary presented. This is part of our hook - “Most grads pick by the bigger number, but our data says the bigger number is the wrong metric. Consider how much you would save…” Other than the default view, the viewer can choose to randomize other cities/salaries or can manually alter the values.

In this way, the map now serves to ground the viewer in our scenario/message: you are a new grad with a lot of opportunity to relocate ahead of you, but how do you make the decision?

